ANNEX 1 (part A)

Bio-based Industries Research and Innovation action

NUMBER — 745695 — VALUEMAG

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1.1. The project summary

Project Number [1]	745695	Project Acronym [2]	VALUEMAG

One form per project

General information

Project title [3]	Valuable Products from Algae Using New Magnetic Cultivation and Extraction Techniques
Starting date [4]	The first day of the month after the signature by the JU
Duration in months [5]	36
Call (part) identifier [6]	H2020-BBI-JTI-2016
Topic	BBI-2016-R09 Exploiting algae and other aquatic biomass for production of molecules for pharma, nutraceuticals, food additives and cosmetic applications
Fixed EC Keywords	Marine biorefineries, Bioproducts (products that are manufactured using biological material as feedstock) biomaterials, bioplastics, biofuels, bioderived bulk and fine chemicals, bio-derived novel materials
Free keywords	
Abstract [7]	

VALUEMAG project aims to provide groundbreaking solutions for microalgae production and harvesting as well as scaling up biomass transformation systems in order to provide new technologies for aquatic/marine biomass integrated bio-refineries. Production-cultivation and harvesting objectives are achieved by using magnetic nanotechnologies: superparamagnetic iron oxide nanoparticles (SPAN) are introduced into microalgae protoplasm in order to confer them magnetic properties. Magnetic microalgae (MAGMA) are immobilized onto a soft magnetic conical surface (SOMAC) and covered with a thin layer of continuously circulating water. A greenhouse hosts SOMAC system to exposure MAGMA to sunlight, minimize contamination and temperature-humidity uncertainties. Quantity of water is minimized and harvesting will be fast and inexpensive. These innovations permit optimum cultivation, enhance biomass productivity and dramatically lower costs of biomass production. Biomass is directly utilized by VALUEMAG multi-facilities bio-refinery for the production molecules for pharmaceutical, nutraceuticals, food additives and cosmetics. Using selected microalgae strains, natural products will be extracted by supercritical CO2 extraction, while a new selective magnetic separation method for precise selection of value-added products will be also developed. To reduce the amount of greenhouse gases realized in the environment and further lower costs of biomass, CO2 produced by transformation processes as well as water are recycled and used to enhance microalgae growth rate. All together VALUEMAG achievements will perfectly meet the demand of capturing the potential of aquatic biomass. The project outputs will bring to the market a broad variety of value-added products in sustainable way. Finally, competitiveness of the European industry will be improved since there are not pilot installations or stateof-the-art bio-refineries utilizing magnetic nanotechnology to cultivate microalgae.

1.2. List of Beneficiaries

Project Number [1]	745695	Project Acronym [2]	VALUEMAG

List of Beneficiaries

No	Name	Short name	Country	Project entry date[8]	Project exit date
1	NATIONAL TECHNICAL UNIVERSITY OF ATHENS - NTUA	NTUA	Greece		
2	UNIVERSITA DEGLI STUDI DELLA CAMPANIA LUIGI VANVITELLI	SUN	Italy		
3	AGENZIA NAZIONALE PER LE NUOVE TECNOLOGIE, L'ENERGIA E LO SVILUPPO ECONOMICO SOSTENIBILE	ENEA	Italy		
4	NOMASICO LTD	NOMASICO	Cyprus		
5	THERACELL ADVANCED BIOTECHNOLOGY LTD	THERACELL	United Kingdom		
6	FYZIKALNY USTAV SLOVENSKEJ AKADEMIE VIED	FU SAV	Slovakia		
7	PNO INNOVATION	PNO	Belgium		
8	INNOVACIO I RECERCA INDUSTRIAL I SOSTENIBLE SL	IRIS	Spain		
9	EXERGY LTD	EXERGY	United Kingdom		
10	VERTECH GROUP	VERTECH	France		
11	ECODUNA AG	ECODUNA	Austria		the day after the notification of termination
12	EPARELLA GMBH	EPARELLA	Austria	01/09/2017	

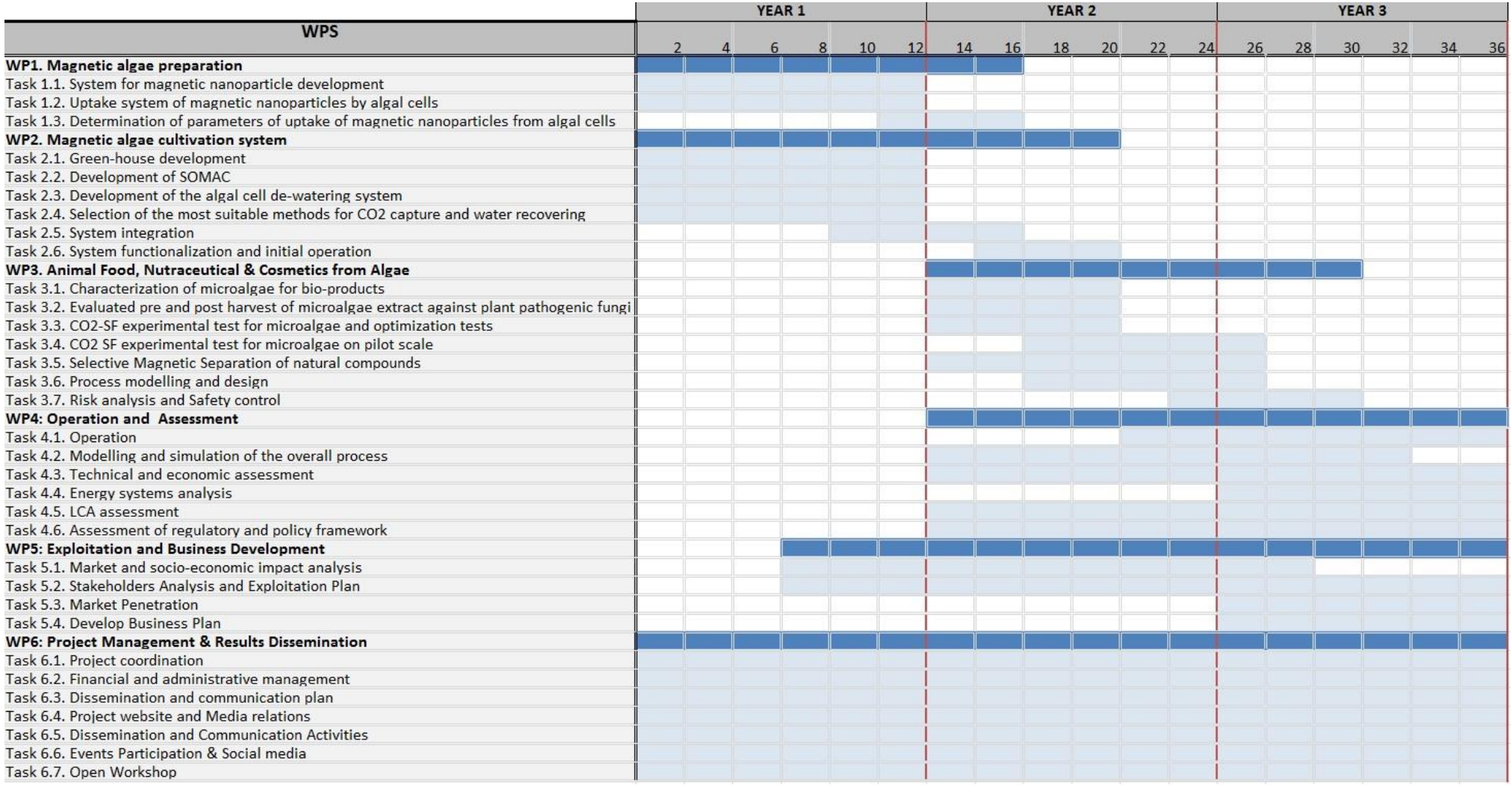
1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	WP Title	Lead beneficiary[10]	Personmonths[11]	Start month[12]	End month[13]
WP1	Magnetic Algae Preparation	1 - NTUA		1	16
WP2	Magnetic Algae Cultivation System	4 - NOMASICO		1	20
WP3	Animal Food, Nutraceuticals & Cosmetics from Algae	3 - ENEA		13	30

WP4	Operation and Assessment	2 - SUN		13	36
WP5	Exploitation and Business Development	7 - PNO		6	36
WP6	Project Management & Results Dissemination	1 - NTUA		1	36
		Total			

chart


YEAR 1
YEAR 2
YEAR 3
WPS
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36
WP1. Magnetic algae preparation
Task 1.1. System for magnetic nanoparticle development
Task 1.2. Uptake system of magnetic nanoparticles by algal cells
Task 1.3. Determination of parameters of uptake of magnetic nanoparticles from algal cells
WP2. Magnetic algae cultivation system
Task 2.1. Green-house development
Task 2.2. Development of SOMAC
Task 2.3. Development of the algal cell de-watering system
Task 2.4. Selection of the most suitable methods for CO2 capture and water recovering
Task 2.5. System integration
Task 2.6. System functionalization and initial operation
WP3. Animal Food, Nutraceutical & Cosmetics from Algae
Task 3.1. Characterization of microalgae for bio-products
Task 3.2. Evaluated pre and post harvest of microalgae extract against plant pathogenic fungi
Task 3.3. CO2-SF experimental test for microalgae and optimization tests
Task 3.4. CO2 SF experimental test for microalgae on pilot scale
Task 3.5. Selective Magnetic Separation of natural compounds
Task 3.6. Process modelling and design
Task 3.7. Risk analysis and Safety control
WP4: Operation and Assessment
Task 4.1. Operation
Task 4.2. Modelling and simulation of the overall process
Task 4.3. Technical and economic assessment
Task 4.4. Energy systems analysis
Task 4.5. LCA assessment
Task 4.6. Assessment of regulatory and policy framework
WP5: Exploitation and Business Development
Task 5.1. Market and socio-economic impact analysis
Task 5.2. Stakeholders Analysis and Exploitation Plan
Task 5.3. Market Penetration
Task 5.4. Develop Business Plan
WP6: Project Management & Results Dissemination
Task 6.1. Project coordination
Task 6.2. Financial and administrative management
Task 6.3. Dissemination and communication plan
Task 6.4. Project website and Media relations
Task 6.5. Dissemination and Communication Activities
Task 6.6. Events Participation & Social media
Task 6.7. Open Workshop

1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Production of magnetic nanoparticles by CVD and flame pyrolysis	WP1	1 - NTUA	Report	Public	12
D1.2	System for uptake and separation of magnetic nanoparticles	WP1	4 - NOMASICO	Demonstrator	Public	12
D1.3	Report on operating parameters for uptake and separation of magnetic nanoparticles	WP1	6 - FU SAV	Report	Confidential, only for members of the consortium (including the Commission Services)	16
D2.1	Cultivation chamber development	WP2	4 - NOMASICO	Report	Confidential, only for members of the consortium (including the Commission Services)	15
D2.2	SOMAC development	WP2	1 - NTUA	Report	Confidential, only for members of the consortium (including the Commission Services)	17
D2.3	De-watering system	WP2	12 - EPARELLA	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D2.4	CO2 and Water re-use	WP2	2 - SUN	Report	Confidential, only for members of the consortium (including the Commission Services)	14
D2.5	Integrated system	WP2	4 - NOMASICO	Demonstrator	Public	18

D2.6	System functionalization	WP2	4 - NOMASICO	Report	Confidential, only for members of the consortium (including the Commission Services)	20
D3.1	Characterization of microalgae for bioproducts	WP3	3 - ENEA	Report	Public	20

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D3.2	Report on pre and post harvest of microalgae extract against plant pathogenic fungi	WP3	3 - ENEA	Report	Public	20
D3.3	Report on bench scale CO2-SF investigation and optimization tests	WP3	3 - ENEA	Report	Public	20
D3.4	Report on experimental tests on pilot scale for cosmetics and animal food production	WP3	3 - ENEA	Report	Public	26
D3.5	Report on magnetic separation pilot scheme	WP3	1 - NTUA	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D3.6	Report on process modelling and CO2-SF Unit design	WP3	9 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D3.7	Risk analysis and safety report on CO2-SF Unit design	WP3	3 - ENEA	Report	Public	30
D4.1	Modelling and simulation of the overall process	WP4	9 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	32

D4.2	Result on Energy and Exergy systems analysis	WP4	9 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D4.3	Sustainable microalgae supply potential for Nutraceutical, Cosmetics & Animal Food	WP4	2 - SUN	Report	Public	36
D4.4	Environmental impact assessment and statement for VALUEMAG plant	WP4	2 - SUN	Report	Public	36

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D4.5	LCA for VALUEMAG tech	WP4	10 - VERTECH	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D4.6	Analysis of European and international regulation and evaluation of economic incentives for viability of VALUEMAG plant in comparison with conventional tech	WP4	2 - SUN	Report	Public	36
D5.1	Stakeholder Analysis	WP5	7 - PNO	Report	Public	13
D5.2	Final Exploitation Plan	WP5	7 - PNO	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D5.3	Report on market penetration	WP5	4 - NOMASICO	Report	Public	36
D5.4	Business Plan	WP5	4 - NOMASICO	Report	Confidential, only for members of the consortium (including the Commission Services)	36

D6.1	Project website	WP6	7 - PNO	Websites, patents filling, etc.	Public	3
D6.2	Dissemination and Communication plan	WP6	7 - PNO	Report	Confidential, only for members of the consortium (including the Commission Services)	10
D6.3	Dissemination and Communication material	WP6	7 - PNO	Websites, patents filling, etc.	Public	12
D6.4	Dissemination and Communication reports (I)	WP6	7 - PNO	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D6.5	Open Workshop	WP6	5 - THERACELL	Other	Public	34

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D6.6	Dissemination and Communication reports (II)	WP6	7 - PNO	Report	Public	36

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	Lead beneficiary [10]	1 - NTUA
Work package title	Magnetic Algae Preparation		
Start month	1	End month	16

Objectives

- Development of magnetic nanoparticles by flame pyrolysis in large quantities (1 kg per day or batch) with +10 ppmuncertainty of size and stoichiometry
- Determination of parameters of uptake of magnetic nanoparticles from algal cells
- Controlled uptake of magnetic nanoparticles by algae, using electroporation in large scale

Description of work and role of partners

WP1 - Magnetic Algae Preparation [Months: 1-16]
NTUA, NOMASICO, FU SAV, IRIS, EPARELLA
Magnetic nanoparticles will be developed by flame pyrolysis, in other words using the microelectronics technology for chemical vapour deposition (CVD), the consortium will obtain the development of mass quantities of magnetic nanoparticles of any composition-stoichiometry, any diameter from 5 nm to 0.1 m
Selected microalgae strains will be purchased from international biobanks and dedicated laboratory units will be created at NTUA and NOMASICO laboratories in order to optimize the transformation parameters. To reduce risks connected with culture loss, for each strain two series of stocks are retained, one which supplies the starter cultures for the production system and the other which is only subjected to the handling necessary for maintenance. Researches and technicians from both organizations will be responsible for stock cultures and growth of microalgae species. Pilot cultures will be realized for each strain in order to produce high quantity of microalgae cells that will be used for transformation experiments. Based on the growth properties of microalgae, they will be constantly cultivate in big laboratory flasks in standard conditions (as required for each strain).
For electroporation experiments the following protocol will be used, though it will be adjusted based on the microalga strains:
- Microalgae cells from a fresh culture will be collected and resuspended in approprieted culture medium containingSPANs.
- 100 µl of the cell suspension (ca. 2 x 106 cells) are then transferred to a 0.4-cm-gap cuvette.
- Electroporation pulses will be experimental established.
- Excess of SPANs will be removed by filtration.
- Microalgal cells suspension will diluted with fresh medium and cultured at optimal condition for 1-5 hour.
- Viability of microalgae cells will be measured by counting cells at specific time point for a time-course of 72 hours.The MAGNAs preparation with the described protocol requires only few minutes to transform millions of microalgae cells. However, for large scale application a new kind of elctroporation apparatus will be fabricated in the frame of VALUEMAG that will guarantee high transformation performance. Once the best transformation parameters are set, the introduction of SPANs into microalgae cells will be performed directly at the greenhouse that hosts the mPBR using the new apparatus.

For glass bead preparation the following protocol will be used, also in this case it may be adjusted based on the microalga strains:
- Microalgae cells from a fresh culture will be collected and resuspended in approprieted culture medium containingSPANs.
- Glass beads with a diameter <1 mm will be sterilized by autoclave before to be used.
- Microalgae cells (2-4 x 106 cells) will be suspended in the appropriate medium containing glass beads and SPANs- The suspension will be vortex for 30 seconds and left at room temperature for 30 minutes in order to allow deposition of glass beads at the bottom of the tube.
- Excess of SPANs will be removed by filtration.
- Microalgal cells suspension will diluted with fresh medium and cultured at optimal condition for 5-8 hour.
- Viability of microalgae cells will be measured by counting cells at specific time point for a time-course of 72 hours.

Task 1.1: System for magnetic nanoparticle development. (NTUA, FU SAV) (M1-M12)
In this task, NTUA will develop the methodology, equipment and all procedures in order to produce magnetic nanoparticles with controlled size and stoichiometry. The thermal oxidation or chemical vapour deposition facility of NTUA will be employed, with two minor upgrades: the installation of a precise mass flow supply to the output of a gas mixture fire. The parameters for mass-flow delivery are from 0.1 l/min to 60 l/min, while the output fire is from 700oC

to 2700oC. Parametric control of the used precursors, their mass-flow transmission and the temperature at the point of treatment will result in proper nanoparticle development. Initial characterization by means of XRD, SEM and DTA is to be done at the NTUA.
Consequently, samples will be delivered to FU SAV, in order to characterize them properly. FU SAV will employ XRD for phase determination, DTA for phase transformation determination and STEM for the study of the microstructure of the magnetic nanoparticles. Consequently, together with NTUA, FU SAV will provide proper ab-initio and multi-scale modelling, thus targeting total quality control in the corresponding production.
As above mentioned, the target of Task 1.1 is the development of mono-dispersed magnetic nano-particles of uncertainty in size and stoichiometry less than +10 ppm.

Task 1.2: Uptake system of magnetic nanoparticles by algal cells. (NOMASICO, NTUA, EPARELLA) (M1-M12)
Uptake of magnetic nanoparticles will be realized mainly by electroporation. The technology will be implemented at the premises of NOMASICO with the help of NTUA. The solution of water, microalgal cells and SPANs will be driven to parallel conducting plates, which operate under the influence of electric field. Pulsed electric shocks (power and durations of strokes will be experimentally determined) will opened pores into cell-membrane of microalgae and the SPANs will pass through these pores. Then, the mixture of microalgal cells and not internalized magnetic nanoparticles will be driven through a mild magnetic field gradient in order to separate free magnetic nanoparticles and magnetic algae cell (algal cells fed with magnetic nanoparticles). Such system warrantees that algal cells under-going this procedure are fed with uniform number of magnetic nanoparticles.
The electroporation instrument will be equipped with a magnetic mass monitoring system, developed by the NTUA, to visualize the amount of magnetic nanoparticles in micro-algal cells. In order to achieve monitoring and feedback of all micro-algal cells, polymer tubes of sub-mm diameter will be employed.
At this point the stock culture of MAGNA cells is left for a certain period of time to recover from the electroporation stress in appropriated tanks. After recovering MAGNA cells will be injected in the mPBR in ordered to be circulated and spread onto magnetic surface.

Task 1.3: Determination of parameters of uptake of magnetic nanoparticles from algal cells. (FU SAV, NTUA, NOM ASICO, IRIS) (M11-M16)
FU SAV will evaluate all field experiments by performing structural characterization by means of STEM, XRD and DTA analysis. All data will be evaluated together with NOMASICO and NTUA and corrective actions will be decided to feed the work of the previous Task. Thus, the existing electroporation system will be integrated and optimized during this task, resulting in a determination of uptake parameters with respect to time and electric power provided.

Participation per Partner

Partner number and short name	**WP1 effort**
1 - NTUA	
4 - NOMASICO	
6 - FU SAV	
8 - IRIS	
12 - EPARELLA	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D1.1	Production of magnetic nanoparticles by CVD and flame pyrolysis	1 - NTUA	Report	Public	12

D1.2	System for uptake and separation of magnetic nanoparticles	4 - NOMASICO	Demonstrator	Public	12
D1.3	Report on operating parameters for uptake and separation of magnetic nanoparticles	6 - FU SAV	Report	Confidential, only for members of the consortium (including the Commission Services)	16

Description of deliverables

D1.1 Production of magnetic nanoparticles by CVD and flame pyrolysis, M12
D1.2 System for uptake and separation of magnetic nanoparticles, M12
D1.3 Report on operating parameters for uptake and separation of magnetic nanoparticles, M16

D1.1 : Production of magnetic nanoparticles by CVD and flame pyrolysis [12]
Production of magnetic nanoparticles by CVD and flame pyrolysis

D1.2 : System for uptake and separation of magnetic nanoparticles [12]
System for uptake and separation of magnetic nanoparticles

D1.3 : Report on operating parameters for uptake and separation of magnetic nanoparticles [16] Report on operating parameters for uptake and separation of magnetic nanoparticles

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS2	Operating flame pyrolysis and production of nanoparticles	1 - NTUA	12	Working Demonstrator / initial characterization of nanoparticles
MS3	Transformation of microalgae with nanoparticles	1 - NTUA	12	Viability demonstration and growth property characterization
MS7	Reports	1 - NTUA	36	Reports on time

Work package number [9]	WP2	**Lead beneficiary [10]**	4 - NOMASICO
Work package title	Magnetic Algae Cultivation System		
Start month	1	**End month**	20

Objectives

The main objective of WP2 is the development of the pilot plant system (a single SOMAC will be developed) devoted for magnetic algae cultivation. To obtain it the following objectives are required:
• Development of the cultivation chamber to host the pilot plant
• Development of the soft magnetic cone
• Development of the de-watering system of the produced micro-algal cells
• CO2 capture & water recovering and introducing into the mPBR system
• Integration of all algae cultivation systems in one
• Initial functionalization of the integrated system

Description of work and role of partners

WP2 - Magnetic Algae Cultivation System [Months: 1-20]
NOMASICO, NTUA, SUN, ENEA, FU SAV, IRIS, EPARELLA
The system will be developed and integrated in the premises of NOMASICO, in Cyprus. This decision is based on the climatic conditions of the Island, offering advantages with respect to the other Mediterranean countries involved in the project (Italy, Greece and Spain). Of course, the implementation of the method and the system in other countries like those mentioned above is certain.

Task 2.1: Cultivation chamber development. (NOMASICO, IRIS) (M1-M17)
The development of the cultivation chamber will be realized in the premises of NOMASICO. The size of the cultivation chamber will be limited to a square area of 20 m X 20 m and height of 10 m, hosting a single SOMAC unit of 6 m diameter and ~6 m high. The cultivation chamber will have all required automation to obtain proper conditions of temperature and humidity in the whole volume of it. Targeted temperature and humidity are 25-30oC and 60% relative humidity with uncertainty of +3oC and 8% RH respectively. IRIS will design and develop all required automation system for the cultivation chamber. The cultivation chamber will use CO2 produced by extraction processes, therefore a connection system that drives CO2 from the gasification unit will be design. In addition it will be included a pipeline scheme that will allow to introduce and uptake water as well as biomass. Since microalgae need natural solar radiation the infrastructure will be constructed with transparent or semi-transparent material that allow transmission of electromagnetic radiation in the range of 400-700 nm.
IRIS will design and develop an integrated control system for automating the management and monitoring of the cultivation chamber PBR. This system will include data fusion and intelligence tools such as fuzzy logics for a holistic control of the most important parameters for algae growth and optimal production conditions. The system design will be developed in Deliverable 2.1 as part of the cultivation chamber development.
This Task will include permission of operation from the State and therefore its duration will be 17 months from the beginning of the project.

Task 2.2: Development of SOMAC. (NTUA, NOMASICO, FU SAV) (M1-M15)
The SOMAC will be developed by the NTUA in collaboration with NOMASICO and delivered to the premises of NOMASICO. The cone will be developed from mild steel using welding methods. Stress test and annihilation will be provided after welding due to a unique method of the NTUA. The formation of the surface will be realized by NOMASICO, using their own novel electromagnetic forming method. According to this technology, the steel will be hammered electromagnetically in the place of use, by using a portable and automated system. The SOMAC will host a pipeline scheme that drives water on top of the cone. The SOMAC slope will be design in order to avoid turbulence on water flow that may destabilize the MAGMA cell monolayer. Thickness of water will be adjustable and uniform all over the SOMAC surface. Sensors for temperature and water flow will be included.

Task 2.3: Development of the algal cell de-watering system. (EPARELLA, SUN, ENEA) (M1-M12)
The de-watering system will be developed by the EPARELLA in collaboration with SUN and ENEA. The system will be a closed conduit system and will be parallel connected to several SOMAC schemes. Since the expected microalgal biomass concentration will high 2-3 g/l a filtration unit that concentrate biomass up to 30-40% slurry will be couples with mild heating process based on solar thermal energy. Even though filtration is considered an expensive method to harvest microalgal biomass, it must be say that this is true only when high volume of water must be treated. The key factor is the ratio volume-to-surface. When this ratio is extremely high like in raceways (V/S>=200 l/m2), or in conventional tubular photobioreactors (V/S=100 l/m2) the volume needed per 1 kg of biomass are 4.5 m3/kg and 2.2 m3/kg, respectively. The consequence of this is that the low yields of reactors with high V/S ratio imply that larger volumes of reactors are needed to produce the same amount of biomass, and in consequence the cost of the harvasting can be higher than for low V/S reactors. The very low V/S ratio of reactors based on SOMAC technology (<3 l/m2) together with high growth rate if immobilized MAGNA make affordable filtration costs. It is estimate that the volume needed for 1 kg of biomass is 0.2-0.3 m3.

Task 2.4: Selection of the most suitable methods for CO2 capture and water recovering for the integration with mPBR process, (SUN, ENEA, EXERGY) (M1-M12)
CO2 and water residual integrations with mPRB process will reduce the environmental impact of the VALUEMAG project. The evaluation of the water cleaning and CO2 capturing according the kind of algae will be carried out in parallel and taking into account both process as a whole. According to the results generated throughout task 2.3 and WP3, the best methods will be selected for the VALUEMAG project purpose. The selection of the technologies will be strongly depending on different factors: technological, economic, environmental and legislative which will evaluate the feasible of the technologies; the performance over the different algae system; legislative issues; technical risks; environmental impact (CO2 captured, impact on water remediation). A combination of different methods and the most innovative solutions will be also studied. Moreover the evaluation of water and nutrient savings through aqueous residual cleaning and recycling will be performed. Finally, micro-algae based CO2 bio-mitigation capacity will be analysed and potential emission bio-mitigation effect will be evaluated with respect to micro-algae derived products.

The innovation proposed in VALUEMAG project is to promote the reuse of all organic residues (deriving from other processes of food production, pharma molecules and cosmetics).

Task 2.5: System integration. (NOMASICO, NTUA, SUN, ENEA, IRIS) (M9-M18)
The integration of the whole system, including the system for the development of micro-algal cells, the system for uptake and separation of the magnetic nanoparticles to
micro-algal cells, the SOMAC and the de-watering process will be realized by NOMASICO, with the aid of NTUA, SUN, ENEA and IRIS.

Task 2.6: System functionalization and initial operation. (NOMASICO, NTUA, SUN, ENEA, IRIS) (M15-M20)
The system will be tested as a whole after its integration. Initial corrective actions for operational reasons will be considered after the system integration. During this period, all system have to operate properly, with the supply chain well operating between partners. This sub-task is aimed at being finalized within 6 months.
After that initial functionalization, the complete system will be operated for another 6 months in order to produce repeatedly micro-algae and perform operational parametric adjustments.

The system will be integrated as a small scale production line of micro-algae in Cyprus, concerning all involved parameters and systems, except the production of magnetic nanoparticles, which will remain in NTUA, Greece for several practical reasons.

Participation per Partner

Partner number and short name	**WP2 effort**
1 - NTUA	
2 - SUN	
3 - ENEA	
4 - NOMASICO	
6 - FU SAV	
8 - IRIS	
12 - EPARELLA	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D2.1	Cultivation chamber development	4 - NOMASICO	Report	Confidential, only for members of the consortium (including the Commission Services)	15
D2.2	SOMAC development	1 - NTUA	Report	Confidential, only for members of the consortium (including the Commission Services)	17

D2.3	De-watering system	12 - EPARELLA	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D2.4	CO2 and Water re-use	2 - SUN	Report	Confidential, only for members of the consortium (including the Commission Services)	14
D2.5	Integrated system	4 - NOMASICO	Demonstrator	Public	18
D2.6	System functionalization	4 - NOMASICO	Report	Confidential, only for members of the consortium (including the Commission Services)	20

Description of deliverables

D2.1 Cultivation chamber development, M17
D2.2 SOMAC development, M15
D2.3 De-watering system, M12
D2.4 CO2 and Water re-use, M14
D2.5 Integrated system, M18
D2.6 System functionalization, M20

D2.1 : Cultivation chamber development [15]
Cultivation chamber development

D2.2 : SOMAC development [17]
SOMAC development

D2.3 : De-watering system [12]
De-watering system

D2.4 : CO2 and Water re-use [14]
CO2 and Water re-use

D2.5 : Integrated system [18]
Integrated system

D2.6 : System functionalization [20]
System functionalization

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS4	Operating soft magnetic cone by trapping micro-algae cells	4 - NOMASICO	20	Proper cultivation of microalgae cell without transferring SPAN to new born cells
MS7	Reports	1 - NTUA	36	Reports on time

Work package number [9]	WP3	Lead beneficiary [10]	3 - ENEA
Work package title	Animal Food, Nutraceuticals & Cosmetics from Algae		
Start month	13	End month	30

Objectives

This work package focuses on the valorisation of bio-products from micro-algae by CO2-SF process and by a new experimental technology named Selective Magnetic Separation.
The research will provide:
• Cosmetics & nutraceuticals production with Supercritical CO2 extraction by selected microalgae
• Animal Food production with Supercritical CO2 extraction by selected microalgae
• Scaling up of CO2-SF technology for bio-products (Cosmetics and Animal food by microalgae)• Enhancement of natural products purity and quality by Selective Magnetic Separation

Description of work and role of partners

WP3 - Animal Food, Nutraceuticals & Cosmetics from Algae [Months: 13-30]
ENEA, NTUA, SUN, THERACELL, FU SAV, EXERGY, EPARELLA
Task 3.1: Characterization of microalgae for bio-products (ENEA, FU SAV) (M13 to M20)
Biochemical activity and pharmacological properties of polyunsaturated fatty acids such as eicosapentaenoic acid (EPA, C20:5 -3), docosahexaenoic acid (DHA, C22:6 -3) and the - linolenic acid (ALA, C18:3 -3) are well known. All these fatty acids (n-3 PUFAs) are synthesized from single-celled marine plants, i.e. microalgae, rich in oils with different amounts of -3 fatty acids.
Solvent extraction for the fractionation and characterization of polyunsaturated fatty acids from microalgae -3 (qualitative-quantitative analysis using both the spectrophotometric method and the HPLC), considering the oxidative processes connected, will be carried out, upon completion of the lipids extraction with organic solvents.
The experimental procedure for the complete analysis of lipid materials from microalgae biomass provides:
• the extraction of total lipids through the use of selective solvents;
• the quantification of total lipids by spectrophotometric tests;
• the determination of the qualitative-quantitative composition in unsaturated fatty acids by HPLC.
Similarly, as for the lipid fraction, the extraction and the analysis of the qualitative-quantitative composition for the recovery of astaxanthin will be carried out.

Task 3.2: Evaluated pre and post harvest of microalgae extract against plant pathogenic fungi (ENEA) (M13 to M20)
The main points of this taskwill be:
1) biologically active compounds will be extracted by algal biomass with different processes (solvent extraction,hydrodistillation and fluid supercritical extraction);
2) all the extracts will be tested to evaluate in vitro their inhibition effect on the conidia germination and mycelialgrowth. The best results of biopesticides production will be tested in vivo ; 3) the best extracts will be analysed by HPLC or GAS- Mass;
4) protocols for application of the extracts in vivo will be developed.

Task 3.3: CO2-SF experimental test for microalgae and optimization tests (ENEA, EPARELLA) (M13 to M20)
The extraction with supercritical fluids, widely investigated in several research centres as an emerging technology, is particularly attractive for the use of supercritical CO2 from vegetable matrices for obtaining active ingredients/ biomolecules in the food sector, cosmetics and pharmaceuticals.
The use of supercritical fluids as extraction solvents allows in many cases to lower times, costs and risks compared to traditional techniques with liquid solvents.
For the extraction of natural products, the supercritical CO2 is the preferred solvent since it offers the following advantages:
• a mild process conditions (its critical temperature 31.1 ° C is almost environmental);
• recognized as safe for food products;
• cost effective;
• readily available;
• non-toxic, non-flammable and inert to most materials.

Its superiority compared to conventional technologies of extraction, particularly for natural products in food and pharmaceutical industry, it is now well recognized, due to the possibility to obtain:

- delicate and fresh extracts as natural products;
- active ingredients with high activity;
- extracts with higher life period;
- no residual solvent;
- high flexibility in operating conditions;
- precise control of the selectivity of the supercritical solvent;
- easy to optimize the quality and yield of the desired product;• simultaneous fractionation of extracts.

The use of a non-toxic solvent such as supercritical CO2 makes the extract and all the processing waste totally free of pollutants.

For the development of the optimal n-3 PUFA extraction conditions it is necessary to take into account many factors, such as the particle size of the sample, eventual matrix pre-treatment, the pressure and the temperature (both of which extraction of separation), the flow rate of the solvent and the total time of extraction, as well as the amount of loaded material, the height of the solid filler in the extractor, the extraction mode and the possible use of entrainer co-solvent. The high number of operating variables that affect the extraction process implies that its realization is not immediate and the optimization of the process is complex; to develop a SFE is necessary, in fact, a thorough analysis to study to affirm the feasibility.

The starting point is the searching of the optimal extraction conditions using a laboratory scale plant, and carrying on the investigation in a pilot plant and the operational conditions optimization (which are mainly focused on the influence of the solvent content and its flow rate on the entire process).

Concomitantly extraction with organic solvents will conduct, to evaluate yield and the selectivity of the extraction with supercritical CO2.

The qualitative and quantitative characterization of the extract will be carried out by spectrophotometric analysis for the determination of total lipids and also by HPLC in order to assess both the yield and the selectivity in respect of the unsaturated fatty acids.

Similarly, as for the lipid fraction, it will be used supercritical fluids, CO2 SFE, as an extraction solvent for the astaxanthin recovery. Scientific literature shows that the maximum total of astaxanthin recovery exceeds 97%, calculated from its initial content.

Task 3.4: CO2 SF experimental test for microalgae on pilot scale (ENEA, SUN) (M17 to M26)

A more ambitious goal, considering the high selectivity of CO2 SFE against both the (n-3 PUFA) and the recovery of astaxanthin, would be the construction of predictive models of the extraction process. This would make it possible to extend the validity of test results from the laboratory tests, the scale-up of pilot scale. Unfortunately, although the significant development of SFE applications, there are fragmentary experimental evidences in the literature that show that it can be used for modelling studies.

Task 3.5: Selective Magnetic Separation of natural compounds (NTUA, ENEA) (M13 to M26)

The first step of this task is the identification of surfactants and ligands to selectively separate and purify a given natural molecule. As above mentioned the technology proposed to concentrate compounds is based on magnetic beads covered with a special organic surface able to bind single molecule with high affinity and specificity.

Development of a pilot scheme is the second step. During this phase a main machine (SEPARATOR) will be set up and tested. All components will be purchase, or home-fabricated, and assembled. Initial analytical runs will be performed and the purity of product tested.

Task 3.6: Process modelling and design (EXERGY, ENEA) – (M17 to M26)

Ad hoc software will be developed based on experimental test on pilot CO2-SF scale in order maximize the concentration of some species of interests of the VALUEMAG project, unique calculation tools will be defined along with the exchange of the necessary information. A comprehensive study about existing possibilities will be carried on in order to identify previous developments and implemented solutions and to agree on a common starting point among all the Partners. Finally, outcomes of the simulation will be used to define a preliminary data sheets, as well as the lay-out, drawings and P&IDs of the integrated plant with a technical manual for the same plant with the detected critical issues, that will constitute the starting point for the activities planned in WP4.

Task 3.7: Risk analysis and Safety control (ENEA, EXERGY, THERACELL) (M23 to M30)

This task will be related to the investigation of the potential risk of the designed CO2-SF unit, according to the wellknown risk analysis procedures The system will be assessed and after the definition of the Risk Matrix, the risk reducing measures will be investigated as preliminary tool for the evaluation of severity and frequency of accident. According to this analysis, coupled with the design best practices referred to chemical process control the unit control system will be also designed and tested in a simulative environment. The preliminary evaluation of auxiliary operations units will be completed.

Expected Results

The significant improvement of knowledge in the specific features of CO2-SF of microalgae, in terms of process conditions and reactor characteristics. Materials assessment, safety procedures and design of CO2 -SF reactor for algae processing. Introducing a new magnetic-based technique to improve purity of natural products Optimization process for high value added products production from algae wastes

Participation per Partner

Partner number and short name	WP3 effort
1 - NTUA	
2 - SUN	
3 - ENEA	
5 - THERACELL	
6 - FU SAV	
9 - EXERGY	
12 - EPARELLA	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Characterization of microalgae for bioproducts	3 - ENEA	Report	Public	20
D3.2	Report on pre and post harvest of microalgae extract against plant pathogenic fungi	3 - ENEA	Report	Public	20
D3.3	Report on bench scale CO2-SF investigation and optimization tests	3 - ENEA	Report	Public	20
D3.4	Report on experimental tests on pilot scale for cosmetics and animal food production	3 - ENEA	Report	Public	26
D3.5	Report on magnetic separation pilot scheme	1 - NTUA	Report	Confidential, only for members of the consortium (including the Commission Services)	26

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.6	Report on process modelling and CO2-SF Unit design	9 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D3.7	Risk analysis and safety report on CO2-SF Unit design	3 - ENEA	Report	Public	30

Description of deliverables

D3.1 Characterization of microalgae for bio-products, M20
D3.2 Report on pre and post harvest of microalgae extract against plant pathogenic fungi, M20
D3.3 Report on bench scale CO2-SF investigation and optimization tests, M20
D3.4 Report on experimental tests on pilot scale for cosmetics and animal food production, M26
D3.5 Report on magnetic separation pilot scheme, M26
D3.6 Report on process modelling and CO2-SF Unit design, M26
D3.7 Risk analysis and safety report on CO2-SF Unit design, M30

D3.1 : Characterization of microalgae for bio-products [20]

Characterization of microalgae for bio-products

D3.2 : Report on pre and post harvest of microalgae extract against plant pathogenic fungi [20]

Report on pre and post harvest of microalgae extract against plant pathogenic fungi

D3.3 : Report on bench scale CO2-SF investigation and optimization tests [20]

Report on bench scale CO2-SF investigation and optimization tests

D3.4 : Report on experimental tests on pilot scale for cosmetics and animal food production [26]

Report on experimental tests on pilot scale for cosmetics and animal food production

D3.5 : Report on magnetic separation pilot scheme [26]

Report on magnetic separation pilot scheme

D3.6 : Report on process modelling and CO2-SF Unit design [26]

Report on process modelling and CO2-SF Unit design

D3.7 : Risk analysis and safety report on CO2-SF Unit design [30]

Risk analysis and safety report on CO2-SF Unit design

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS5	Production of food additives	3 - ENEA	26	Achieved objectives (cost & quality)
MS7	Reports	1 - NTUA	36	Reports on time

Work package number [9]	WP4	**Lead beneficiary [10]**	2 - SUN
Work package title	Operation and Assessment		

Start month	13	End month	36

Objectives

This work package focuses on the evaluation and environmental impact assessment
The research will provide:

- To design of fully integrated industrial process incorporating the bioreactors and downstream processes.
- To carry out an economic assessment including market potentials, generation costs related to conventional microalgaeproduction, employment and wealth effects as well as to the economic risks;
- To assess the possible sustainable supply of micro-algae production in EU, suitable for bio-product To investigatehow a large-scale implementation of VALUEMAG technology will affect the surrounding the cosmetics and animal food market;
- To elaborate a complete LCA of different technologies and to design alternatives of the technology;
- Analysis of national and international regulation and assessment of necessary economic incentives for economicviability of microalgae plants

Description of work and role of partners

WP4 - Operation and Assessment [Months: 13-36]
SUN, NTUA, ENEA, NOMASICO, THERACELL, FU SAV, IRIS, EXERGY, VERTECH, EPARELLA
Task 4.1: Operation (NOMASICO, IRIS, EPARELLA, FU SAV, NTUA) – (M21 to M36)
In this task the integrated system will be used as a demonstrator in the place of installation in order to validate it under real conditions and optimize the operation

Task 4.2: Modelling and simulation of the overall process (EXERGY, IRIS) –(M13 to M32)
This task aims to demonstrate the integration of the VALUEMAG process at industrial scale. The starting point for this task will be the data collected from the previous work, including lab scale process and up scaled process results. Once all the data is collected, a virtual up-scale will be carried out by using flow sheet simulation software (AspenOne and/ or SuperPro Designer). Mathematical, empirical and data driven models will be updated and used for this task, with the aim of obtaining analytical information about the real industrial process, requirements and costs. The design will address a fully integrated industrial process incorporating all stages:
- Pre-treatment for the different wastewater streams of the bioreactor.
- The algae cultivation system.
- CO2 capture method.
Other processes like auxiliary services production (heat, steam, cooling water, pressure, etc.) will be also taken into account at this stage. Equipment and reactors for the base case, thermal and energetic processes, ancillary equipment, volume of production, quantity and quality of products, requirements for the industrial plant, energy needs, sub-products and characteristics, etc., will be the output of this Task which will be used for Task 6.2.

Task 4.3: Technical and economic assessment (SUN, ENEA) – (M13 to M36)
Preliminary assessment of potential markets in the world for cosmetics, animal food and pharma compounds. This assessment will be conducted taking into consideration the availability of energy resources, energy supply characteristics and energy demand requirements in the location under investigation, comparing them with conventional technologies, regulatory and policy framework regarding the promotion of renewable bio-product sources and any other circumstances that may affect the deployment of VALUEMAG tech. Preliminary economic assessment of the generation of bioproducts in VALUEMAG-biomass plant. This involves an estimation of capital costs, operating costs and Levelised Cost of Cosmetics & Animal Food for different plant configurations, distribution and utilization strategies; the results would be compared with conventional technologies.
Assessment of employment and wealth effects associated with the deployment of different configurations VALUEMAG plants, its risks associated with future of bio-products price development, component price uncertainty, development of competing technologies and geographical conditions will be assessed.

Task 4.4: Sustainable waste biomass supply potentials (SUN, VERTECH) – (M25 to M36)
Using existing studies, organic waste and agro-industrial biomass will be quantified in order to evaluate the potential WS and at the same time define a possible scenario of short chain on provincial scale in order to apply the same technology

used VALUEMAG project. These data will be intersected with the information about VALUEMAG project at the end to evaluate the sustainability of this application. Energy use and GHG emission from bio-products by microalgae and agroindustrial organics residue will be calculated, separately for biomass production/collection, its production all based on the results of VALUEMAG project in comparison with the BAT available. Carbon, nitrogen and phosphor balances of the integrated process of different microalgae and agro-industrial biomass types will be quantified. A comparison between the tech VALUEMAG and traditional tech will be carried out at the end to evaluate the potential of VALUEMAG respect to the state of art.

Task 4.5: Energy and exergy systems analysis (EXERGY) – (M13 to M36)
Energy system analysis will reveal effects for surrounding energy system on: electricity prices, system costs and CO2 emissions. Existing model will be used as basis and further model development will be carried out skilled for the VALUEMAG tech..
Exergy analysis is a well-established and wide used methodology to evaluate the real efficiency of energy-intensive industrial processes and the entire plant. Through this analysis, the quality of energy and material streams in the process (in terms of work equivalents) are expressed. This task will study process life cycles from a thermodynamic perspective in order to analyse inefficiencies in process chains and propose improvement potential. An exergy assessment methodology to determine specific exergy contents to all mass and energy in and outflows will be developed. When necessary, computerized simulation tools (i.e. ASPEN or TRNSYS) will be utilized.
The potential for improvements will be assessed by the overall exergy efficiency, which is the real exergy over the exergy used in the process. This study will compare the efficiency with regards to other similar processes. To achieve this, indicators can be used which present a reasonable approach in communicating performance results in an objective way. Aside from the determination of the efficiency for the different streams, exergy analysis will establish the potential causes for the detected inefficiency and will decrease cost associated to operation and maintenance of the plant by means of a suitable assignment of energy and material flow and measures the contamination level of the process. Additionally, the effect of construction and operation options on exergy consumption will also be investigated and resource strategies will be derived from the results of the exergy analysis

Task 4.6: LCA assessment (VERTECH) – (M13 to M36)
Firstly, a series of simplified LCA will be carried out in order to investigate the environmental performance and carbon footprint of alternative ENEA plant configurations and operating strategies. The studies will evaluate the effect of microalgae sourcing, plant configuration operating strategies and scale, location and production. Inventory data for these analyses will be obtained primarily from standard databases and published papers. The results will be compared against conventional plants for Animal Food & Cosmetics. These results will be considered for the design and selection of operating conditions and strategy of the DEMO.
Standard LCA factors (ReCiPe, EPD 2007, CED) will be used to evaluate the impact on specific environmental categories, primarily global warming, depletion of non-renewable resources, eutrophization and ecotoxicity. A series of modifications to the original design and strategy will be suggested in order to optimize the environmental performance of the plant. SimaPro8 will be used to produce the LCA models and computational requirements.

Task 4.7: Assessment of regulatory and policy framework regarding process and products (SUN, THERACELL) – (M25 to M36)
This task will involve a survey of national and international regulations affecting the construction and operation of VALUEMAG plants in locations of interest. This may include an survey of national energy strategies, international obligations regarding emission of greenhouse gases, existing and expected regulations regarding the promotion of renewable energies for bio-products and its use in Cosmetics & Animal Food sectors. In addition, a more specific analysis of energy and environmental policy and regulations will be carried out and used in the decision making process. The level of economic support required to make this technology economically viable and attractive to investors will be assessed considering both the estimated present and future prices on conventional bio-products for Cosmetics & Animal Food fields.

Expected Results
- Determination of the potential global market penetration of VALUEMAG including the competitiveness of thetechnology;
- Determination of the effects on the surrounding energy system as a result of implementation of VALUEMAG technology;
- An overview over present relevant national and international regulation and adjustments needed for the viability ofthe VALUEMAG technology.

Participation per Partner

Partner number and short name	WP4 effort

1 - NTUA	
2 - SUN	
3 - ENEA	
4 - NOMASICO	
5 - THERACELL	
6 - FU SAV	
8 - IRIS	
9 - EXERGY	
10 - VERTECH	
12 - EPARELLA	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D4.1	Modelling and simulation of the overall process	9 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D4.2	Result on Energy and Exergy systems analysis	9 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D4.3	Sustainable microalgae supply potential for Nutraceutical, Cosmetics & Animal Food	2 - SUN	Report	Public	36
D4.4	Environmental impact assessment and statement for VALUEMAG plant	2 - SUN	Report	Public	36
D4.5	LCA for VALUEMAG tech	10 - VERTECH	Report	Confidential, only for members of the consortium (including the Commission Services)	36

D4.6	Analysis of European and international regulation and evaluation of economic incentives for viability of VALUEMAG	2 - SUN	Report	Public	36

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
	plant in comparison with conventional tech				

Description of deliverables

D4.1 Modelling and simulation of the overall process, M32
D4.2 Result on Energy and Exergy systems analysis, M32
D4.3 Sustainable microalgae supply potential for Nutraceutical, Cosmetics & Animal Food, M36
D4.4 Environmental impact assessment and statement for VALUEMAG plant, M36
D4.5 LCA for VALUEMAG tech, M36
D4.6 Analysis of European and international regulation and evaluation of economic incentives for viability of VALUEMAG plant in comparison with conventional tech, M36

D4.1 : Modelling and simulation of the overall process [32]

Modelling and simulation of the overall process

D4.2 : Result on Energy and Exergy systems analysis [32]

D4.2 Result on Energy and Exergy systems analysis

D4.3 : Sustainable microalgae supply potential for Nutraceutical, Cosmetics & Animal Food [36]

Sustainable microalgae supply potential for Nutraceutical, Cosmetics & Animal Food

D4.4 : Environmental impact assessment and statement for VALUEMAG plant [36]

Environmental impact assessment and statement for VALUEMAG plant

D4.5 : LCA for VALUEMAG tech [36]

LCA for VALUEMAG tech

D4.6 : Analysis of European and international regulation and evaluation of economic incentives for viability of VALUEMAG plant in comparison with conventional tech [36]

Analysis of European and international regulation and evaluation of economic incentives for viability of VALUEMAG plant in comparison with conventional tech

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS6	Production of nutraceuticals & pharma molecules	2 - SUN	26	Successful characterization of products
MS7	Reports	1 - NTUA	36	Reports on time

Work package number [9]	WP5	**Lead beneficiary [10]**	7 - PNO
Work package title	Exploitation and Business Development		
Start month	6	**End month**	36

Objectives

This WP aims at:
- Developing an exploitation strategy to foster continuation of the project results (networks, services and Intelligenceand Partnering Platform) after the project ends. Achieve significant exploitation impact;
- Stimulate market penetration
- Developing a high level business plan

Description of work and role of partners

WP5 - Exploitation and Business Development [Months: 6-36]
PNO, NTUA, SUN, ENEA, NOMASICO, THERACELL, FU SAV, IRIS, EXERGY, VERTECH, EPARELLA
Task 5.1 - Market and socio-economic impact analysis (PNO, EXERGY, VERTECH) M6-M26
The market aspects (PNO) of the products will be analysed via market prices, market potential, and main factors influencing the future market development. The results are compared to the conventional reference products based on a cost/revenue consideration. Additionally some key parameters from socio-economic aspects (LCA) e.g. job creation, effects on GDP, import/export will be analysed based on the whole value chains.

Task 5.2 - Stakeholders Analysis and Exploitation Plan (PNO, ALL) M6-M36
In the pursuit of sustainable impact, the VALUEMAG Consortium will deliver an Exploitation Plan, identifying exploitable results by each of the partners in the Consortium, types of exploitation and the most suitable platforms for each type of exploitation. In order to determine for specific stakeholders what type of Exploitation actions add most value and what message content should be used, a systematic stakeholder analysis relying on a proven methodology developed by PNO will identify the most important stakeholder groups within and around the VALUEMAG value chain and assess their position towards the project's results in order to set up engagement strategies. The methodology contemplates :
• The identification of the value chains that benefit from the project results based on a joint brainstorm discussion withthe partners.
• The identification of relevant stakeholders belonging to the identified value chains based on the exploration of differentdatabases such as:
o The client portfolio and network of the consortium partners that can be interested in the developed technologies o The CORDIS database where innovators and potential end users will be identified o Patent databases where innovators and potential investors and end users will be identified o Recent scientific publications and reports where innovators and industrial partners will be identified o Specific websites and press releases related to the project topic where potential investors will be identified
Relevant stakeholders will be interviewed or invited to participate in a survey, which will be designed to measure stakeholder characteristics, like their interest, attitude, influence and knowledge relevant for the project. Based on the outcome of the interview/survey, more targeted dissemination and exploitation actions will be implemented and relationships will be built with key stakeholders to the project.

Task 5.3– Market Penetration (NOMASICO, EPARELLA, THERACELL, IRIS) M25-M36
Market intelligence and Private Equity will be searched from the beginning of the project in order to have the investment needed for construction of the minimum viable production plant for producing and selling the first batches of product with the least possible risk, in line with the concept of distributed plants (close to raw materials) that we want to set for these biorefineries. For such a goal, investment ready materials (investment pitch, financial projections) will be prepared and current contacts with European networks of business angels and corporate ventures will be multiplied.

Task 5.4 – Develop Business Plan (NOMASICO, THERACELL, ALL) M25-M36
The following works will be carried out during the development of a business plan for the VALUEMAG technology: Develop a coherent approach to the market sector business model; Final products Value; Qualify the required investment; Qualify risks associated with market entry, investment and applications' development; Draft road map; Model marketing definition including start-up costs, licences, permits, etc.; Breakeven analysis; Definition of risk management strategies

Participation per Partner

Partner number and short name	WP5 effort
1 - NTUA	
2 - SUN	

3 - ENEA	
4 - NOMASICO	
5 - THERACELL	
6 - FU SAV	
7 - PNO	
8 - IRIS	
9 - EXERGY	
10 - VERTECH	
12 - EPARELLA	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D5.1	Stakeholder Analysis	7 - PNO	Report	Public	13
D5.2	Final Exploitation Plan	7 - PNO	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D5.3	Report on market penetration	4 - NOMASICO	Report	Public	36
D5.4	Business Plan	4 - NOMASICO	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Description of deliverables

D5.1 Stakeholder Analysis, M13
D5.2 Final Exploitation Plan, M36
D5.3 Report on market penetration, M36 D5.4 Business Plan, M36

D5.1 : Stakeholder Analysis [13]
Stakeholder Analysis

D5.2 : Final Exploitation Plan [36]
Final Exploitation Plan

D5.3 : Report on market penetration [36]
Report on market penetration

D5.4 : Business Plan [36] Business Plan

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS7	Reports	1 - NTUA	36	Reports on time

Work package number [9]	WP6	Lead beneficiary [10]	1 - NTUA
Work package title	Project Management & Results Dissemination		
Start month	1	End month	36

Objectives

This WP aims at:
Project Management
To keep the project on track and on time; to find solutions for any deviations and minimize the delays to project plans; to manage the interaction among the partners and contacts with the EC.
Dissemination of the Results
- Making an inventory of relevant communication & dissemination channels and target groups, leveraging on earlier;
- Caring out continuous dissemination activities throughout the project duration, according to the dissemination plan;- Ensuring the widespread dissemination of the projects results to stakeholders, industry and general public through the use of multiple communications tools, online media, media relations, workshops and events.
- Developing an exploitation strategy to foster continuation of the project results (networks, services and Intelligenceand Partnering Platform) after the project ends. Achieve significant exploitation impact; - Coordinating the overall communication, dissemination strategy of the project.

The VALUEMAG Consortium will use a variety of communications platforms to reach the relevant stakeholders and maintain communication with them. That will include the creation and maintenance of a project website, establishing presence in a number of social media networks in order to reach the wider public and increase the project's public awareness and carefully selected participation in events organised by others. Furthermore, partners will have presence at various BioEconomy and circular economy trade shows. The final workshop will be the focal point for attracting stakeholders outside the consortium.

Description of work and role of partners

WP6 - Project Management & Results Dissemination [Months: 1-36]
NTUA, SUN, ENEA, NOMASICO, THERACELL, FU SAV, PNO, IRIS, EXERGY, VERTECH, EPARELLA
Task 6.1: Project coordination (NTUA) M1-M36
Coordinate the project activities, encourage efficient interaction between the partners, manage the risks, organize meetings and provide documents and reports to the EC.

Task 6.2: Financial and administrative management (NTUA, ALL) M1-M36
Distribute funds, monitor and manage the project's expenses, manage IPR issues, prepare the Consortium Agreement to be submitted and then signed by all partners.

Task 6.3: Dissemination and communication plan (PNO, ALL) M1-M36
At the start of the project, the Consortium will review, map and expand the list (email pool) of SMEs, other stakeholders related to the project innovation clusters, institutions, organisations, groups, projects, experts, stakeholders and develop a dissemination plan (a Working Document) with tailored means to reach main target audiences. During this phase, PNO will also gather partners' input, descriptions, available messaging and logos, photos, visuals to be used wherever needed in the project tasks. The dissemination plan will also include the project key messages and project Q&A to be modified accordingly to fit the targeted stakeholder groups.
Project results will also be presented to relevant specialised National and international conferences throughout the duration of the project. The partners will share among them the most suitable symposia, conferences and workshops, based on the information they already have in hand. The most appropriate among these conferences will be identified and mapped during the elaboration of the Dissemination Plan. Potential reach to key stakeholders, opportunities for feedback, timing, and impact of proceedings will be among the criteria for selecting a particular conference for the project partners to participate.

Task 6.4: VALUEMAG website and Media relations (PNO) M1-M36
In the beginning of the project, WP8 will set up the VALUEMAG website. The website, in English, will be developed to provide an overview of the VALUEMAG objectives and programme of work, to track its progress and present relevant results. Project deliverables will be available on the website and project workshops and activities will be promoted. The website will be publicly available and will serve as a useful source of information for the relevant audience and other stakeholders. Partners will have a dedicated page on the project website.
The website will also include a blog, providing opportunities for comments and discussions on main project themes. It will also advertise a dedicated e-mail address, e.g. info@VALUEMAGproject.eu that can be used by interested parties

to express their views and make inquiries to the consortium. Once an inquiry has been made, it will be forwarded to the most suitable expert in the Consortium.
The main goal of media relations is to raise the project's profile and news in the international media on an ongoing basis. In this context, this WP would draft and disseminate press releases and would seek publication opportunities to ensure the project's basic information and results coverage. Furthermore, project partners will also explore opportunities to publish findings in specialised journals, identified in the elaboration of the Dissemination Plan.

Task 6.5: Dissemination and Communication Activities (PNO) M1- M36
The first step to a successful project dissemination and communications strategy and implementation is the visual identity of the project itself. A VALUEMAG logo and style guide will be designed and produced. The logo and the style guide will be used on two A4 or A5 leaflets, the website, one poster and two project exhibition pop-up banners that provide the wider audience with the key project messages. The leaflets will serve as a leave-behind at conferences and events and provide the different audiences with the key project information. The poster and the pop-up banners are necessary exhibition marketing tools. A project video will be produced during the second half of the project.
Infographics are a popular marketing tool nowadays. They outline complex processes and issues in simple pictures that are easy to comprehend. The project infographic will serve as an extra marketing tool. Part of the communications marketing materials will also be a Power Point Presentation template with the VALUEMAG logo on, the project's key messages and main description, to be used by the partners on each occasion needed
A project newsletter will be produced on a five basis (five newsletters in total) presenting results of the different work packages. The newsletter will be circulated to project partners and a large variety of stakeholders based on the email pool that will be put together at the beginning of the project. The newsletter will be available to download on the VALUEMAG website.
Additionally, VALUEMAG project partnership will create and manage a dedicated communication campaign supporting VALUEMAG output as competitive with other biobased products and materials, in order to foster the market penetration of the VALUEMAG technology. This communication campaign will be focusing on industry stakeholders, but also research community and potential consumers.
Particular attention will be paid to the engagement of Classification Societies, such as DNV-GL to adopt the idea, as well as the local coastal Society of Cyprus.

Task 6.6: Events Participation & Social media (PNO, ALL) M1 – M36
Based on the events identified in the initial communications plan, the Consortium members will attend, present and discuss the development of the VALUEMAG project, boosting its visibility to the rest of the relevant community.
Dissemination and communication towards policy makers, EC representatives and international stakeholders will be provided by organising VALUEMAG events in Brussels. In 2019 VALUEMAG partners will organise the "VALUEMAG Open Workshop in Brussels".
Another channel for dissemination of the VALUEMAG objectives, organisation, and results is through social media. This WP will set-up and maintain a presence in social media networks, including LinkedIn, Facebook and Twitter. The VALUEMAG social media presence will broaden the reach to the public and experts interested in following the project and it development and offering feedback and suggestions.
The presence in social media, along with the website, will allow to make a rough estimate on how innovative, substantiated and valuable views are expressed by consortium members and outside stakeholders, e.g. by the number of hits/visits, 'likes' etc.

Task 6.7: Open Workshop (THERACELL, ALL) M1-M36
The main findings of the project will be presented at a final open workshop. This event will also provide opportunities for policy makers and other stakeholders to share their views and comment on the project results, and will thus directly enhance the marketability of the VALUEMAG project. Workshop will take place in Brussels. The event will involve external speakers and participants in order to present VALUEMAG project results in the European

Participation per Partner	
Partner number and short name	**WP6 effort**
1 - NTUA	
2 - SUN	
Partner number and short name	**WP6 effort**

3 - ENEA	
4 - NOMASICO	
5 - THERACELL	
6 - FU SAV	
7 - PNO	
8 - IRIS	
9 - EXERGY	
10 - VERTECH	
12 - EPARELLA	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D6.1	Project website	7 - PNO	Websites, patents filling, etc.	Public	3
D6.2	Dissemination and Communication plan	7 - PNO	Report	Confidential, only for members of the consortium (including the Commission Services)	10
D6.3	Dissemination and Communication material	7 - PNO	Websites, patents filling, etc.	Public	12
D6.4	Dissemination and Communication reports (I)	7 - PNO	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D6.5	Open Workshop	5 - THERACELL	Other	Public	34
D6.6	Dissemination and Communication reports (II)	7 - PNO	Report	Public	36

Description of deliverables

D6.1 Project website, M3
D6.2 Dissemination and Communication plan, M10
D6.3 Dissemination and Communication material, M12
D6.4 Dissemination and Communication reports (I), M18

D6.5 Open Workshop, M34
D6.6 Dissemination and Communication reports (II), M36

D6.1 : Project website [3]
Project website

D6.2 : Dissemination and Communication plan [10]
Dissemination and Communication plan

D6.3 : Dissemination and Communication material [12]
Dissemination and communication material

D6.4 : Dissemination and Communication reports (I) [18]
Dissemination and Communication reports (I)

D6.5 : Open Workshop [34]
Open Workshop

D6.6 : Dissemination and Communication reports (II) [36] Dissemination and Communication reports (II)

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS1	Kick-off meeting	1 - NTUA	1	Minutes of meeting
MS7	Reports	1 - NTUA	36	Reports on time

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Kick-off meeting	WP6	1 - NTUA	1	Minutes of meeting
MS2	Operating flame pyrolysis and production of nanoparticles	WP1	1 - NTUA	12	Working Demonstrator / initial characterization of nanoparticles
MS3	Transformation of micro-algae with nanoparticles	WP1	1 - NTUA	12	Viability demonstration and growth property characterization
MS4	Operating soft magnetic cone by trapping micro-algae cells	WP2	4 - NOMASICO	20	Proper cultivation of microalgae cell without transferring SPAN to new born cells
MS5	Production of food additives	WP3	3 - ENEA	26	Achieved objectives (cost & quality)
MS6	Production of nutraceuticals & pharma molecules	WP4	2 - SUN	26	Successful characterization of products
MS7	Reports	WP1, WP2, WP3, WP4, WP5, WP6	1 - NTUA	36	Reports on time

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	One or more partners leave the consortium	WP6	Effective management procedures are applied to timely intercept problems, remove or reallocate partners and, if needed, replace them with new partners with suited skills and profiles. The Consortium Agreement will realise the policies behind this.
2	Partners fail to communicate effectively	WP6	Discuss the problem individually with the concerned partners. Discuss the situation at the technical management and at the General Assembly level. Find a way to communicate that all partners can accept. Provide tools and utilities to facilitate successful communication and collaboration.
3	Disability of producing magnetic nanoparticles (MN)	WP1	Buying magnetic nanoparticles from other flame pyrolysis manufacturers or laboratories, keeping trying to produce them in the lab

4	Disability of introducing MN in the algae cells	WP1	Use higher intensities of electric field at different frequencies; simultaneous use of temperature elevation for the enlarging the algae cell pores; use alternative glass beads method.
5	Split of magnetic nanoparticles to new-born cells during cell duplication	WP1, WP2	Control the amount of MN introduced in average in the algae cells, in terms of time exposure to the electric field; change amplitude and frequency of electroporation; change temperature during electroporation; pre-selection of collected new born cells with magnets
6	Disability of SOMAC to trap MAGMA	WP2	Re-design of the surface of the soft magnetic cone; re-design of the polarizing electromagnets (set inside the SOMAC volume)
7	Disability to produce nutraceuticals, food additives and pharma products	WP3	Use magnetic techniques by means of covering magnetic particles or nanoparticles with proper surfactants attracting those nutraceuticals

1.3.7. WT7 Tentative

to be provided.

4. Members of the consortium

4.1. Participants (applicants)

Partner 1 - NTUA	National Technical University of Athens School of Electrical and Computer Engineering Laboratory of Electronic Sensors www.ece.ntua.gr/en

Organisation Description

The National TU of Athens is the leading TU of Greece, with 9 Schools, 550 Staff and 25,000 students. It is one of the leading organizations in Europe concerning competitive projects.

The Laboratory of Electronic Sensors, National Technical University of Athens, Greece, is active on magnetism, magnetic materials and their applications in sensors and transducers, as well as in other electronic systems. The lab is well experienced in magnetic materials and their applications to sensors and systems, as well as in designing new instrumentation for testing and characterization. The team is involved in the manufacturing and treatment of bulk materials, thin films, rapidly quenched materials, powders etc, by using planar flow casting, chemical vapour deposition, film oxidation, physical vapour deposition, electrodeposition, wet milling, annealing & field annealing and stress-current annealing techniques. It also performs structural characterization, by means of high resolution transmission electron microscopy and scanning electron microscopy, x-ray diffraction, X-ray photoemission spectroscopy, atomic and magnetic force microscopy, as well as properties determination using alternating gradient field magnetometry, magnetostriction measurements, magnetotransport properties, ac susceptibility and all of them with parametric control of frequency, temperature and stress.

The lab is involved in projects related to the design of sensors and sensor systems for various applications. All these activities have resulted in smart specialization for commercial products. The group is strongly related with local and European enterprises concerning the transformation of the prototypes developed at the laboratory to pre-industrial products and final products,

In the recent 10 years, the group is active in energy and health projects related to magnetically enhanced methods for bio-fuels and renewable fuels, such as hydrogen production from water, bio-diesel production from wasters etc., as well as bio-engineering applications, such as hyperthermia, nanoparticle monitoring in the body, production of phenols from Greek agricultural products etc.

Relevant Publications

1. ND Papadopoulos, HS Karayianni, PE Tsakiridis, M. Perraki, E. Sarantopoulou, E. Hristoforou, MOCVD Cobalt Oxide Deposition from Inclusion Complexes: Decomposition Mechanism, Structure, and Properties, Journal of the Electrochemical Society, 158, p. 5-13, 2011.
2. G.S.E. Antipas, E. Statharas, P. Tserotas, N. Papadopoulos and E. Hristoforou, Experimental and first principles' characterization of functionalized magnetic nanoparticles, ChemPhysChem, 14 (9) , pp. 1934-1942, 2013
3. DN Kossyvakis, CG Vossou, CG Provatidis, Computational and experimental analysis of a commercially available Seebeck module, Renewable Energy 74, pp. 1-10, 2014
4. GSE Antipas, E Mangiorou, E. Hristoforou, The effect of indium content on the atomic environment and cluster stability of GeSe4Inx=10,15 glasses, Metals, 5 (1), pp. 102-118, 2015
5. DN Kossyvakis, CG Vossou, CG Provatidis, EV Hristoforou, Computational analysis and performance optimization of a solar thermoelectric generator, Renewable Energy, 81, pp. 150-161, 2015

Relevant Projects

- Magnasense, Clean Sky project, with NTUA partner
- Inducer, Clean Sky project, with NTUA partner
- Several other national, European and International project including industrial subcontracting
- HELENIC-REF, FET-OPEN REA Project, with NTUA coordinator

Infrastructure/Equipment

Process

- Complete thick film production line for hybrid structures
- Thermal oxidation, chemical vapour deposition and flame pyrolysis
- Physical vapour deposition (sputtering, electron beam, thermal evaporation)
- Microelectronic technology (wire bonders, wedge bonders, assembly) Rapidly quenched ribbons **Characterization**
- High resolution transmission electron microscope JEOL 2100
- Field emission gun – scanning electron microscope with e-beam lithography
- X-ray photoemission spectroscopy
- X-ray diffraction with Euler Cradle set-up
- Magnetic characterization set-up **Applications**
- Laser interferometer for position calibration
- Mass flow meter, pressure and balance calibrators
- Magnetometer calibrator
- Temperature and humidity calibrators
- Stress calibration in magnetic steels

Profile of Main Persons

Evangelos V Hristoforou (male), D.Eng, Ph.D, C.Eng, is Professor of Electronic Materials at the School of Electrical and Computer Engineering, and Director of the Laboratory of Electronic Sensors, National Technical University of Athens (NTUA). His group of electronic and magnetic materials (www.magnetics.metal.ntua.gr) at the NTUA is performing research in Materials and Applications for Sensors and Transducers (MAST), with the most important product concerning a method and instrument to monitor the 3-d stess tensor distribution in steels, in Hybrid Electric Energy Integrated Cluster (HELENIC), with the most important advances referring to the sustainable and affordable production of hydrogen from water and Magnetism for Biomedical Applications (MBA), a new-born activity, concerning driving and monitoring of magnetic nanoparticles. He is actively related to several industries in the field of sensors, energy and health, running industrial and research projects. He has published more than 150 papers in international journals (h-index=23) and he gave more than 50 invited talks in international conferences. He is the: Representative of Greece in the Universal Network of Magnetic NDE; ViceChair of the Hellenic NDT Association; Chair of the European Magnetic Sensor and Actuator (EMSA) Conference; Chair of the International Conference on Materials and Applications for Sensors and Transducers (IC-MAST); Editor-in-Chief in "Journal of Materials and Applications for Sensors and Transducers"; Regional Editor in "Recent Patents in Material Science"; Editorial Board in "Journal of Electrical Engineering", "Journal of Automation Mobile Robotics and Intelligent Systems" and "Measurement Automation Robotics".

Dr. Aphrodite Ktena (female) is Associate Professor in the Electrical Engineering Department, TEI of Sterea Ellada, Greece. She received Ph.D. and M.Sc. degrees in Electrical and Computer Engineering from the Carnegie Mellon University, Pittsburgh, PA, USA (1993) and B.Sc. degree in Electrical Engineering from the University of Bridgeport, CT, USA (1989). Her research interests include hysteresis and magnetic materials modelling; characterization of magnetic materials; sensor development and measurement technology; ICT applications in power engineering and education. Over the last decade she has been working closely with the Laboratory of Physical Metallurgy, NTUA, Greece, on magnetic non-destructive testing (MNDT) methods for steels, sensor development, and modelling. Her work has been presented in over 50 ISI journal and international conference publications. She has participated in a multitude of European funded projects and coordinated national funded projects.

Polyxeni Vourna (female), was born in 1982 in Athens, Greece. She received the degree in Physics from the University of Patras, Greece, in 2001 and MSc Degree in Data Communication Systems from Brunel University of London, UK, in 2009. She obtained her second MSc degree in Automated Systems from National Technical University of Athens in 2010. From 2011 till 2015, she was PhD candidate under the supervision of Prof. Evangelos Hristoforou. From October 2015 she serves as Research Associate in the group of electronic and magnetic materials, responsible for structural characterization, properties determination and properties – structure correlation. She has published more than 15 papers in international journals and conferences.

Kata Berkesi (female) MSc, Ph.D has graduated as an Environmental Engineer in 2009 in faculty of radiochemistry at University of Pannonia, Hungary, Veszprém. Her Master thesis was awarded as the best thesis in 2009 among the Hungarian students graduated that year. In 2014 she finished her PhD with the topic of combined radiochemical and electrochemical methods at Doctoral School for Chemical and Environmental Sciences. In 2012 she started as research assistant and finished as research fellow at Wigner Research Centre for Physics of the Hungarian Academy of Sciences in . Her projects included anodized oxidation of alumina-sheet to produce Al_2O_3 porous membranes, electrodeposition of NiP-amorphous coating for neutron-beam guides, electrochemical preparation of Co/Bi magnetic/nonmagnetic multi-layered nanostructures. In 2015 within a framework of a Hungarian-Greek scholarship won for 10 months she started her work under the supervision of Prof. E. Hristoforou at the School of Mining and Metallurgy Engineering at NTUA. Later in 2015 she has joined to HELENIC-REF project of NTUA as a postdoctoral researcher for the sub-project of catalyst production. She has 3 highlighted peer-reviewed publications, 2 of them as a first author. She is reviewer of two international journals.

Spyridon Aggelopoulos MSc (male) is PhD Candidate in the Electrical Engineering Department of the National Technical University of Athens (NTUA). In 2014, he received his MSc degree in Automation Engineering from the Mechanical Engineering Department of the NTUA. In 2012, he received his Bachelor Diploma on Applied Physics from the School of Applied Mathematical and Physical Sciences of the NTUA. His interests include the study, design and development of magnetic sensors and the design, simulation and development of electronic circuits.

Mariliza Kouli MSc (female), is PhD Candidate in the Electrical Engineering Department of NTUA. She received her MSc degree in Automation Engineering from the Mechanical Engineering Department of NTUA (2015). Her Bachelor Diploma on Materials Science and Engineering was received from the University of Ioannina (2013). Her research interests include electromagnetic NDTs on magnetic and composite materials, characterization of low carbon steels and research and development of magnetic nanomaterials.

Christina Stefani MSc (female) is PhD Candidate in Electrical Engineering Department of National Technical University of Athens. In 2015 she received her MSc degree in Physics and Technological Applications from the School of Applied Mathematical and Physical Sciences of NTUA. Her Bachelor Diploma on Physics was received from National Kapodestrian University of Athens in 2013. Her research interests include magnetic materials modelling and electromagnetic detox techniques on iron oxides.

Sofia Papadopoulou (female) attends to the Interdepartmental Program of Postgraduate Studies in Microsystems and Nanodevices, in the School of Applied Mathematical and Physical Sciences of the National Technical University of Athens, since 2015. She received her Bachelor Diploma on Mining and Metallurgical Engineering from NTUA (2015). Her research interests mainly include magnetic sensors and rapidly quenched materials.

Dr. Nikos Stefanakis (male) is currently a researcher at the National Technical University of Athens (NTUA) - Dr. Stefanakis has extensive experience in design, modelling and optimization of engineering processes. Throughout his work he has involved in several European and International projects, developing workable and often innovative solutions, aimed at efficient systems. Key areas of work have included design and analysis of advanced solar thermal collectors, 3rd generation concentrating photovoltaic systems, and adsorption chillers. Before NTUA, he was with Heliotron Energy SA, as a Product & Research Engineer. Since 2002, he has successfully participated in more than 15 R&D and private projects. Dr. Stefanakis received his Diploma in Chemical Engineering (2002), and a PhD in Process Analysis & Plant Design (2009) from NTUA. His research was supported by public (European Social Fund 80% and GSRT 20%) and private funds, within measure 8.3 of Op. Pr. Comp., 3rd CSP-PENED 2003

Partner 2 - SUN	**Second University of Naples** **Department of Civil and Building Engineering, Design and Environment** **Group of Innovative technologies for environment protection from pollution and sustainable resource use** www.dicdea.unina2.it
Organisation Description	

The Second University of Naples (SUN) is one of the leading Italian universities, with 2 Schools, the School of Medical Sciences and the School of Polytechnics and Basic Sciences, 19 Departments and over 30,000 students. SUN participates in national and international organization initiatives and research projects, in order to promote the development and the dissemination of knowledge, contributing to the execution of scientific cooperation and higher education programs, through membership in networks and international consortia.

The Department of Civil and Building Engineering, Design and Environment (DICDEA) of the Second University of Naples (SUN) includes 14 full professors, 14 associate professors, 21 researchers, 28 postdocs, 2 technicians and 37 PhD students. In 2012-2014 the scientific production of DICDEA included over 150 publications in international scientific journals and over 150 conference contributions, with more than 20 funded research projects by private and public authorities.

The research group *Innovative technologies for environment protection from pollution and sustainable resource use – InnoTEP* (**Coordinator: Prof. Dino Musmarra**), of DICDEA, includes scientists in chemistry, environmental science and civil-environmental-chemical engineering and underlines interests in multidisciplinary research topics, focusing on their application in real problems. Mass transfer in gas-solid fluidised beds, gas-liquid absorption, solid dissolution, reaction kinetics, crystallization, adsorption and fluidization have been the main investigation sectors. Moreover, flue gas desulphurization from thermal power plants and micro-pollutant removal from waste incinerator exhaust gas have been studied, by using both catalytic and not catalytic processes. New research themes have been started in the last years, regarding solid waste treatment, reactive streams in unconfined natural environments, groundwater remediation technologies, waste water innovative treatment processes, renewable fuel production, production and testing of advanced adsorbing materials and environmental quality analysis.

Relevant Publications

- Di Natale, F., Erto, A., Lancia, A., Musmarra, D., “Equilibrium and dynamic study on hexavalent chromium adsorption onto activated carbon”. Journal of Hazardous Materials 281: 47-55, 2015.
- Chianese, S., Loipersböck, J., Malits, M., Rauch, R., Hofbauer, H., Molino, A., Musmarra, D., “Hydrogen from the high temperature water gas shift reaction with an industrial Fe/Cr catalyst using biomass gasification tar rich synthesis gas”. Fuel Processing Technology 132: 39–48, 2015.
- Capocelli, M., Prisciandaro, M., Lancia, A., Musmarra, D., “Hydrodynamic cavitation of p-nitrophenol: A theoretical and experimental insight”. Chemical Engineering Journal 254: 1-8, 2014.
- Molino, A., Erto, A., Di Natale, F., Donatelli, A., Iovane, P., Musmarra, D., “Gasification of granulated scrap tires for the production of syngas and a low-cost adsorbent for Cd(II) removal from wastewaters”. Industrial and Engineering Chemistry Research 52(34): 12154-12160, 2013.
- Karatza, D., Lancia, A., Prisciandaro, M., Musmarra, D., Mazziotti Di Celso, G., “Influence of oxygen on adsorption of elemental mercury vapors onto activated carbon”. Fuel 111: 485-49, 2013.
- Chianese, S., Fail, S., Binder, M., Rauch, R., Hofbauer, H., Molino, A., Blasi, A., Musmarra, D., Experimental investigations of hydrogen production from CO catalytic conversion of tar rich syngas by biomass gasification, Catalysis Today, 2015, doi: 10.1016/j.cattod.2016.04.005.
- Di Natale, F., Erto, A., Lancia, A., Musmarra, D., “Mercury adsorption on granular activated carbon in aqueous solutions containing nitrates and chlorides”. Journal of Hazardous Materials 192(3): 1842-1850, 2011.
- Erto, A., Andreozzi, R., Di Natale, F., Lancia, A., Musmarra, D., “Experimental and statistical analysis of trichloroethylene adsorption onto activated carbon”. Chemical Engineering Journal 15: 353-359, 2010.
- Vicinanza, D., Contestabile, P., Ferrante, V. "Wave energy potential in the north-west of Sardinia (Italy)", Renewable Energy, ISSN 0960-1481, vol. 50, pp. 506-521, 2013.

Relevant Projects
• "Hybrid Electric Energy Integrated Cluster concerning Renewable Fuels (HELENIC-REF)" – Fund: 2.578.386,00 Euro (SUN Unit Fund: 553.500 Euro) Call: H2020-FETOPEN-2014-2015-RIA. • "An Innovative Method for Improving the Structural Integrity using SMA Revolutionary Technology (InnoSMART)" – Fund: 1.995.113,00 Euro (SUN Unit Fund: 466.000 Euro). Call: H2020-FETOPEN2014-2015-RIA. • "Mobile Device for Desorption of Hydrocarbons (DI.MO.D.I.)" – Fund: 4.071.907,17 Euro (SUN Unit: 1.254.063,00 Euro). Call: Piano Operativo Nazionale Ricerca e Competitività 2007-2013. • "Environmentally Friendly Coastal Protection in a Changing Climate (EnviCOP)" - Fund: 227.200 euro Call: FP7-PEOPLE-2011
Infrastructure/Equipment
• Fluidized bed pilot plant • Autoclave bench-scale reactor • Vapour mercury monitoring system • Hydrocarbon monitoring system • Gas Chromatography (FID and TCD) • Gas mixing system • COMSOL MultiphysicsTM Licence • CALL PUFFTM Licence
Profile of Main Persons

Dino Musmarra (male), PhD, has been Associate Professor of Chemical Plants of the Department of Civil and Building Engineering, Design and Environment of the Second University of Naples since 2001 and he is the Coordinator of the research group *Innovative technologies for environment protection from pollution and sustainable resource use* (InnoTEP). After graduating as a chemical engineer from the University of Naples "Federico II", he started his career as a researcher in the Combustion Research Institute of the National Research Council in 1986. He received his Ph.D. in Chemical Engineering in 1989 from the University of Naples "Federico II". He is currently teaching in the following courses: Environmental chemical engineering, Pollution reduction technology, Solid waste treatment plants and Remediation of Contaminated Sites. His research interests include: mass and heat transfer; macro- and micro-pollutant removal gaseous streams; flue gas desulphurization from thermal power plants; micro-pollutant removal from waste incinerator exhaust gas; heat transfer in fluidized bed reactors; innovative waste water treatment; water body quality analysis; protection of ground-waters by means of permeable adsorbing barriers; waste water treatment by using Advanced Oxidation Processes; modeling of pollutant in the environment and the environmental quality analysis. During his 25-year research career, he has served as principal investigator in 8 projects and project coordinator in 12 projects, funded with more than €4M for the development of contaminated groundwater remediation technologies, flue gas depuration technologies, renewable energies production technologies and development of innovative materials. He has published more than 100 papers in international journals and more than 60 papers in international conferences. His papers have been hetero-cited more than 1,000 times in journal publications (h-index=26). His research group is currently involved in the *Future Emerging Technologies (FET) – H2020* founded Projects "Hybrid Electric Energy Integrated Cluster concerning Renewable Fuels (HELENIC-REF)" and "An Innovative Method for Improving the Structural Integrity using SMA Revolutionary Technology (InnoSMART)", in which he is Principal Investigator of the SUN Unit. He is reviewer in more than 15 international journals and more than 20 international conferences.

Alessandro Mandolini (male), PhD (1993), has been Assistant Professor (1996-2001), Associate Professor (2001-2011), now Full Professor (2011 to date) of Geotechnical Engineering at the Department of Civil and Building Engineering, Design and Environment of the Second University of Naples. Since 2013 he has been elected Head of the Department. In 2015 ha has been also elected Chair of the Technical Committee TC212 of the International Society of Soli Mechanics and Geotechnical Engineering. His research interests include Foundation Engineering, Earth Retaining Structures, Earthquake Engineering. He has been Partner and Coordinator of several national projects in the last 20 years. Author/Co-author of 121 papers (SCOPUS h-index 8, 267 total citations, 33 documents) on national and international conference and journals. He is reviewer for 8 international journals. He is member of the Editorial Board of the international journal "Geo-Engineering Science".

Diego Vicinanza (male), PhD (1997), has been Assistant Professor (1998-2013) and actually Associate Professor (2014 to date) of Coastal Engineering at the Department of Civil and Building Engineering, Design and Environment of the Second University of Naples. His research interests include coastal engineering, coastal hydrodynamic, wave energy converters. He has been Partner and Coordinator of 18 national and EU projects in the last 20 years. Author/Co-author of 142 papers (SCOPUS h-index 14, 439 citations, 54 documents) on national and international conference and journals. He is reviewer for 24 international journals. He is member of the Editorial Board of Renewable Energy for the subject areas of "Ocean Energy and Offshore Wind".

Pasquale Iovino (male), PhD, has been Assistant Professor of Environmental Chemistry of the Department of Environmental, Biological and Pharmaceutical Sciences and Technologies, of the Second University of Naples since 2011. He received his Ph.D. in Resources and Environment in 2007 from the Second University of Naples. His research interests are related to development and implementation of quantitative determination of exposure markers of air pollution, adsorption of organic pollutants in aqueous solution on activated carbon, application of natural zeolites to water purification from humic substances and emerging contaminants, wastewater treatment by using Advanced Oxidation Processes. He published more than 40 papers on international journals and more than 20 papers in international conference. He is an elected member of the board of the Italian Chemical Society, Section Campania for the year 2015-2017. Furthermore, he is the cofounder of the university spinoff
Environmental Technologies srl

Armando Di Nardo (male), PhD, has been Assistant Professor of Water Infrastructure of the Department of Civil and Building Engineering, Design and Environment of the Second University of Naples since 2010. He graduated in

Civil Engineering from the Second University of Naples in 2001 and received his Ph.D. in Hydraulic and Environmental Engineering from the University of Naples "Federico II" in 2005. His research interests include water resource management optimization, with particular attention to artificial reservoirs, water loss identification and reduction, smart water network sectorization and aquifer remediation, focusing on permeable adsorbing barrier design, developing heuristic optimization technique based on genetic algorithm, graph theory and fuzzy logic. He published more than 100 papers on national and international conference and Journals. He is the Coordinator of the European Action Group of EIP (European Innovation Partnership) on Water, titled Cloud Technologies & ReaL time monitoring + Smart Water Network.

Dr. Despina Karatza (female) graduated in Chemical Engineering from the University of Naples "Federico II" in 1990. Since 1991 she started a collaboration with the Combustion Research Institute of the National Research Council as researcher. Her research interests include flue gas desulphurization from thermal power plants, liquidvapour equilibrium and micro-pollutant removal from waste incinerator exhaust. Her work has been presented in over 35 ISI journal and national and international conference publications. She has participated in a multitude of national funded projects, in particular during her research career, she has served as principal investigator in 6 projects.

Simeone Chianese (male), PhD, he is research fellow at the Department of Civil and Building Engineering, Design and Environment and he belongs to the SUN Units of the InnoSMART Project (funded F.E.T. – H2020) and HELENIC-REF Project (funded F.E.T. – H2020). Graduated cum laude in Environmental Engineering in 2011, at the Second University of Naples, he received his PhD in in Occupational Medicine, Science and Technology for the Environment, confirmed by Postgraduate Program in Science for Life and Environment of the Second University of Naples, with research focusing on the study of syngas CO-shift reaction and syngas cleaning; supervisors: Prof. Dino Musmarra (Second University of Naples), Dr. Reinhard Rauch (Vienna University of Technology) and Dr. Antonio Molino (ENEA – Trisaia Research Centre). His research interests include biomass gasification, catalytic hydrogen production, syngas cleaning, micro-pollutant removal from aqueous streams, advanced wastewater treatment, innovative groundwater remediation by means of permeable adsorbing barriers. His work has been presented in several international scientific journals and international and national conferences.

Partner 3 - ENEA	
ENEA Agenzia nazionale per le nuove tecnologie, l'energia e lo sviluppo economico sostenibile	**Agenzia nazionale per le nuove tecnologie, l'energia e lo sviluppo economico sostenibile** hwww.enea.it
Organisation Description	

ENEA is the name for Italian National Agency for New Technologies, Energy and Sustainable Economic Development and hosts over 2700 staff employees, distributed in its 9 research centers all over the national territory. It conducts research and innovation activities, and provides public administration, enterprises and citizens with its advanced services.

Specifically, ENEA is concerned with energy efficiency, renewable energy sources, nuclear energy. It hosts experimental laboratories and facilities, and also dedicates its technological innovation skills to cultural heritage conservation, agro-food, health, and the environment.

ENEA research activities are focused on the following sectors:

ENERGY EFFICIENCY : Support to Public Administration, Information and Training; Advanced Technologies for Energy and Industry

RENEWABLE ENERGY SOURCES: Concentrated Solar Thermal Energy; Photovoltaics; Biomass and Biofuels; Solar Thermal Energy at low and medium temperatures; Hydrogen, Fuel Cells and Energy Storage Systems

NUCLEAR ENERGY: Nuclear Fusion; Nuclear Fission

CLIMATE AND THE ENVIRONMENT: Environmental Characterization, Prevention and Recovery; Environmental Technologies; Energy and Environmental Modeling; Marine environment and Sustainable Development; Antarctic Expeditions and Research in Polar Areas

SAFETY AND HEALTH: Seismic Protection; Radiation Biology and Human Health; Radiation Protection;

Metrology of Ionizing Radiation

NEW TECHNOLOGIES: Materials Technologies; Radiation Applications; Sustainable Development and

Innovation of the Agro-Industrial System; ICT

ELECTRIC SYSTEM RESEARCH: Studies and research, under a Programme Agreement with the Italian Ministry of Economic Development, aimed at innovating the National Electric System to make it cheaper, safer and more environmentally-friendly.

Besides research, ENEA acts as national Agency in support of central and local Public Administration, enterprises and citizens.

Specifically, ENEA provides:

- Services to Enterprises
- Support to Public Administration
- Technologies and Expertise for Cultural Heritage Preservation
- Radiation Protection Services
- Radioactive Waste Management Services

ENEA promotes collaboration with foreign organizations and institutions in order to participate in research programs and activities of the main international bodies acting in science and technology, also providing expertise for the definition of technical standards.

The main initiatives are implemented within the Agency in the framework of:

- EU ACTIVITIES: EURATOM; Projects funded by the EU, European Technology Platforms; RTD Framework Programme Committees; ENEA Office in Brussels; Strategic Initiatives
- BILATERAL COOPERATION
- MULTILATERAL COOPERATION.

In particular, the Agency's international activities are carried out through:

- participation in projects mainly financed by EU programs, as well as European Technology Platforms (ETPs) and international initiatives;
- support to the activities of its delegates / experts within international R&D committees and groups;
- promotion of the relations with the Ministry of Foreign Affairs, Italian scientific attachés abroad, scientific attachés of foreign embassies in Italy, and the Italian Permanent Representation to the EU;
- reception, at its research centres, of official foreign delegations, also aimed at defining specific bilateral and multilateral collaboration agreements and Memoranda of Understanding;
- organization of international events and meetings;

- hospitality at its laboratories of foreign fellows working in the areas of main interest of the Agency.

ENEA international activities are coordinated and supervised by the International Relation Service (Contact address) ENEA also operates through its ENEA-EU Liaison Office in Brussels , promoting and strengthening the image of the Agency and supporting the development of research projects and activities within the European Union.

Relevant Publications

- Molino A., Migliori M., Macrì D., Valerio V., Villone A., Nanna F., Iovane P., Marino T., Glucose gasification in super-critical water conditions for both syngas production and green chemicals with a continuous process. Renewable Energy, vol. 91; p. 451-455, 2016.
- Molino A., Braccio G., Synthetic Natural Gas SNG production from biomass gasification - Thermodynamics and processing aspects. FUEL, vol. 139; p. 425-429, ISSN: 0016-2361, 2015.
- Molino A., Nanna F, Migliori M, Iovane P, Ding Y, Bikson B., Experimental and simulation results for biomethane production using peek hollow fiber membrane. FUEL, vol. 112; p. 489-493, ISSN: 0016-2361, 2013.
- Molino A., Nanna F, Villone A, Iovane P, Tarquini P, Migliori M, Giordano G, Braccio G., Pressure and time effect over semi-continuous gasification of zootechnical sludge near critical condition of water for green chemicals production. FUEL, vol. 136; p. 172-176, ISSN: 0016-2361, 2014.
- De Corato U., Maccioni O., Trupo M., Di Sanzo G., Use of essential oil of Laurus nobilis obtained by means of a supercritical carbon dioxide technique against post harvest spoilage fungi Crop Protection. Vol. 29, Issue 2, p. 142-147, 2010.
- Ambrico A., Trupo M,. Evaluation of antifungal activity of morphologically distinct colonies of Bacillus subtilis strain ET-1. Journal of Plant Pathology (2011), 93 (4, Supplement), S4.25. Ed. ETS. ISSN: 11254653, 2011.

Relevant Projects

- UNIQUE - Integration of particulate abatement, removal of trace elements and tar reforming in one biomass steam gasification reactor yielding high purity syngas for efficient CHP and power plants, Compact version of a gasifier by integrating the fluidized bed steam gasification of biomass and the hot gas cleaning and conditioning system into one reactor vessel. FP7 (2007-2013), Cooperation, Energy, Collaborative research
- BRISK - The European Research Infrastructure for Thermochemical Biomass Conversion; Networking activities integration to foster a culture of co-operation between the participants in the project, and the scientific communities benefiting from access to the research infrastructures, with the pursuit of joint research activities, and facilitate transnational access by researchers to one or more infrastructures among those operated by participants; FP7 (2007-2013), Capacities, Research infrastructures, Coordination and supporting actions
- TIME - Technological improvement for ethanol production from lignocellulose, Development of cost effective and sustainable production methods for ethanol based on lignocellulosic waste materials or dedicated crops; FP5 (1998-2002), Energy, Environment and Sustainable Development (EESD), RDT
- BIO-H2 - Production of clean Hydrogen fuel cells by reformation of bioethanol, Production of H2 from biomass-derived ethanol, for mobile applications, through processes that offer zero emissions of pollutants; FP5 (1998-2002), Energy, Environment and Sustainable Development (EESD), RDT
- BIOPAL - Algae as raw material for production of bioplastics and biocomposites contributing to sustainable development of EU coastal regions, Design and process a new generation of algae derivated bioplastics and biocomposite targeting applications in agriculture, automotive and packaging industry; biofilter systems for heavy metal accumulation including the biodegradation and biorecycling aspects; FP5 (1998-2002), Quality of Life and management of living resources (Quality of Life), RDT

Profile of Main Persons

Antonio Molino (Male) - Permanent researcher Antonio Molino at the ENEA Portici. PhD Degree in Chemical and Materials Engineering awarded at the University of Calabria. Master degree in Chemical Engineering awarded at the University of Naples Federico II (2003). Master in Project Management (2016). His primary scientific interest and activity is focused on the processes and systems for thermal conversion of waste and biomass, such as pyrolysis and gasification. He is member of several Environmental Committees, co-author of almost 70 papers in scientific journals, most of them peer-reviewed, invited speaker in several conferences. He is currently member of the IEA Bioenergy, National Task Leader 33, Thermal Gasification of Biomass.

Roberto Balducchi (Male) – Senior Researcher at ENEA Italian national agency for new technologies, energy and sustainable economic development. Head of "*Bioproducts and Bioprocesses Laboratory*", in charge for the biotechnologies and agro-industry research, development, dissemination and transfer activities until the scientific and technical support to creation of news biotech *start-up*. He's Member of Working Group "Bioeconomy" of National Ministry of Economic Development; Member of the Working Group "Quality and manufacturing" of Italian Food for Life Platform; Member of various Biological and Biotechnological Associations. He is member of several Environmental Committees, co-author several papers in scientific journals, most of them peerreviewed, invited speaker in several conferences.

Neeta Sharma (Female) - Dr. Sharma has been associated with ENEA (Italian National Agency for New Technologies, Energy and Sustainable Economic Development) for the last more than two decades. Following the successful completion of B.Sc, M.Sc and M.Phil, PhD and Post-Doctoral Fellowship at the University of Milano, at present she has been working as the Responsible for National/International Projects at ENEA. Her personal research expertise includes agro-industrial and bioenergy fields and the biomass production of dedicated energy crops. She has made an excellent research collaboration links with the Stakeholders of national and international reputes including policy makers, Industries and prominent RTD institutes in the field of biobased economy. She is a member of scientific and review committees of various international organizations and has authored over 100 peer-reviewed publications on the sustainable and environmental friendly biomass production of new energy crops for biorefineries. Her expertise and experience has prepared to lead her to the FP7 EU Coordination and support action project entitled "Strengthening networking on Biomass research and biowaste conversion – biotechnology for Europe India integration (SAHYOG)". She organized wide range of international networking events (including project twinning, short-term exchanges of researchers, stakeholder workshops, summer schools), in particular the very first edition of the EU-India Brokerage Event on Bio-economy, to mobilize and stimulate the stakeholders from EU, Member States and India, from both Public and Private sector. She has been the winner of the "Woman in Question", award 2014, dedicated to the women who have made outstanding contributions in projects related to the environmental protection of the territory.

Alfredo Ambrico (Male) - He is permanent researcher at the ENEA Trisaia. PhD Degree in Plant Pathology awarded at the University of Bari (Italy). He is currently working on the control of postharvest diseases of fresh fruit by using essential oils, antagonist microorganisms and ozone as alternative means to synthetic fungicide. The research is documented by articles in journals of scientific interest and participation in conferences of national and international importance. He was coauthor of the Italian patent No. MI2006A000503 "Use of Aphanocladium album Preuss W. Gans as limiting parasitic alterations".

Giuseppe Di Sanzo (Male) - Permanent researcher Giuseppe Di Sanzo at the ENEA Trisaia. Master degree in Chemical Engineering awarded at the University of Calabria His primary scientific interest and activity is focused on the processes and systems for separation system using supercritical fluid as CO2 in supercritical conditions.

Patrizia Casella (Female) – She is a Research Fellow at the ENEA, Research Centre of Portici. PhD degree in Ecology and Management of Biological Resources conferred at University of Tuscia (Italy) (2012). She graduated cum laude in Biology and Ecology of Coastal Marine Environment at University of Messina (Italy) (2008). Her research study is focused on microbiology, studying the role of bacterial communities in natural ecosystems. She has gained extensive experiences on the microbiological techniques for the isolation, the cultivation, the control and the counting of the bacterial cultures. Her scientific production included international scientific papers, proceedings in national and international conferences.

Juri Rimauro (Male) - Researcher at the Environmental Biogeochemistry Laboratory of the Research Center in

Portici (Naples), ENEA (Italian National Agency for New Technologies, Energy and Sustainable Economic Development). Master of Science in Environmental Science with a major in marine environment at the University of Naples "Parthenope." The main scientific interests are focused on ecological and environmental chemical processes in terrestrial and marine environments. The main research activities are sampling of environmental matrix (solid, liquid and gaseous) and chemical qualitative and quantitative analysis of environmental matrix using inductively coupled plasma mass spectrometry (ICP-MS), UV-visible spectrometry, ion chromatography, high performance liquid chromatography (HPLC) and gas chromatography with mass spectrometry (GC-MS). Co-author of many publications in scientific journals, national and international conference proceedings and presentations.

Partner 4 - NOMASICO	**NOMASICO LTD** www.nomasico.com
NomaSico	

Organisation Description

Nomasico LTD of Nicosia, Cyprus is a company mainly involved in the field of surface engineering, machinery manufacturing, generation of novel electromagnetic based techniques for optimisation of structures. It's team of engineers is capable of performing a series of analyses, laboratory experiments and acquisition of data. Specialising in Electrospark Deposition, Electromagnetic Forming (Annihilation of stresses - Forming to specified geometries), Electromagnetic Cutting, FEM, BEM, Discrete Elements, meshless methods, Database / IT tools, namely, matlab SPSS / data analysis, data management systems processing algorithms & tools for large scale data, data mining. Modelling skills including Materials design by modelling , Effective material properties of inhomogeneous materials, Phenomenological thermo-mechanical models of materials response, Micromechanical and multiscale modelling, Damage and fracture, Chemo-mechanical processes and Transport phenomena.

Relevant Publications

1. Real-time monitoring of nanomolar binding to a cyclodextrin monolayer immobilized on a Si/SiO 2/novolac surface using white light reflectance spectroscopy: The case of triclosan
2. Davide Maffeo, Zhivko Velkov, Konstantinos Misiakos, Konstantina Mergia, Anastasia Paulidou, Maria Zavali, Irene M. Mavridis, Konstantina Yannakopoulou. Journal: Journal of Colloid and Interface Science - J COLLOID INTERFACE SCI , vol. 358, no. 2, pp. 369-375, 2011
3. Elucidation of the Local Character of Chemical Reactivity through the Time-Resolved Chromatographic Analysis of Local Molecular Properties of Gaseous Molecules Adsorbed on Solid Surfaces E. D. Metaxa*J. Phys. Chem. C 115, 25389–25412(2011),
4. FRA-1 protein overexpression is a feature of hyperplastic and neoplastic breast disorders. BMC Cancer. Chiappetta G, Ferraro A, Botti G, Monaco M, Pasquinelli R, Vuttariello E, Arnaldi L, Di Bonito M, D'Aiuto G, Pierantoni GM, Fusco A. 2007 Jan 25;7:17.

Relevant Projects

Privately funded projects on valuable products from Greek & Cypriot agricultural products

Infrastructure/Equipment

- Electrospark Deposition
- Electromagnetic Forming
- Electromagnetic Cutting
- Big Data

Profile of Main Persons

Dr. Konstantinos Kalkanis (male) was awarded his Bachelor's degree in Mechanical Engineering (2002) and Masters Degree in Advanced Manufacturing Technology (2003) from the University of Portsmouth. He was awarded his PhD from the National Technical University of Athens (2009). The thesis involved strength of composite materials and health monitoring thereof. He has worked in the construction sector of public works. Further to that he has executed-contributed in tasks in 10 European Union funded scientific programs, with the Greek counterpart being, the National Technological University of Athens and the Technological Institutions of Athens and Ionian Islands. The thematology of tasks included but was not limited to Study of the behaviour of composites under fatigue loading, Structural repairs, experimental and analytic approaches towards implementation thereof, Development of novel materials, Investigation into the generation of smart structures via the incorporation of sensors, Cure cycle examination and certification. He has Lectured from 2005 to 2009 at the National Technical University of Athens at a Post graduate level. He also Lectured from 2004 to now at the Technological Educational Institute of Piraeus and Chalkis in the School of Technical Applications, Sector of Civil Technologists and the Sector of Mechanical Engineers. Author and contributor of 11 published papers in International peer-reviewed scientific journals, and responsible for numerous oral presentations in National and International Conferences. Elected member in the BoD Of the Hellenic Society for Nondestructive Testing from 01/01/09 to 31/12/15 (reelected 3 times). Responsible of Public Relations and the connection of the NDT certification to the Greek educational system.

Dr. Angelo Ferraro (male) in 2002 was awarded by the University of Calabria (Italy) with a degree in Biological Sciences. In 2010 he finished the PhD in Molecular Medicine at the University of Naples "Federico II" (Italy).
Then he moved in Greece and he had worked as post-doctoral research at the National Hellenic Research Foundation (NHRF) for 5 years. Is main field of research is cancer molecular biology. Until now he published 36 peer-reviewed papers 5 of which as first author (total impact factor> 200, h-index=19). He was involved as main researcher in an FP7 (-241783) grant named EpiDiaCan, a Greek-EU project named POM and a second FP7(HEALTH) grant named THALAMOSS. Currently he has a contract of collaboration with the National Technical University of Athens, school of Mining and Metallurgy Engineering (Greece) being responsible for the development and implementation of bio-physics experiments and projects.

Dr. Eleni Metaxa (female) received the Degree of Bachelor in Chemistry (1991) from the Department of Chemistry at the National Kapodistrian University of Athens, Hellas; she was awarded the PhD Degree (1999) from the School of Chemical Engineering at the National Technical University of Athens (NTUA), in the discipline of Chemical Kinetics; she received the Specialization Certificate (2005) in the discipline of Energy Systems Management from the Technological Educational Institute of Piraeus, Hellas, in cooperation with Open University of England, and in 2014 she received her MSc Degree in Catalysis & Protection of Environment from the School of Exact Sciences & Technology of Hellenic Open University, Patras, Hellas. She has served as Teaching Laboratory Assistant at various Hellenic Technological Educational Institutes (2001-2015); she was an external collaborator Chemist (April-September 2002) at the Chemical Laboratory of the Steam-Electric Power Station of Laurium, Hellas; she participated as a Researcher in two Research-Projects (2005-2008) at the School of Chemical Engineering of NTUA and from 2015 to date (2016-) she also participates in the Research-Project HELENIC-REF at the School of Mining and Metallurgy Engineering of NTUA and she was an external collaborator Chemist (2014-2015) at Private Company

“INTERGEO Environmental Technology Ltd”, Thessaloniki, Hellas, active in the field of geo-environmental studies and projects, with particular specialization in polluted soil reclamation and groundwater and hazardous waste management. Her research interests include adsorption and catalysis studies; investigation of sustainable solutions for wastes and synthesis of magnetic nanoparticles for use in environmental and biomedical applications. She has published 18 papers in scientific international journals with high impactfactor, she is one of the four members of a World Patent, she has written a book-chapter after invitation, she has made 10 announcements both in International and National Conferences and she is reviewer in one journal. She is member of the Hellenic Association of Chemists.

Maria Zavali (female) received the Degree in Materials Science and Engineering, School of Sciences and Technologies, University of Ioannina (2006). She was awarded the European Postgraduate Programme on Biomedical Engineering (BME) organized by the Faculty of Medicine of the University of Patras, the Faculty of Mechanical Engineering and the Faculty of Electrical and Computer Engineering of the National Technical University of Athens, Greece. Majoring in Biomedical Informatics and Electronics (2009). From 2009 to present she has worked in the European industry Greece and Germany) and especially as a Design and Analysis Engineer handling Pre-processing, CAD and Weight Management Projects. In research, she has undertaken various tasks including Research Assignment for the National Centre of Scientific Research (N.C.S.R.) “Demokritos” at the Institute of Microelectronics on the subject of: The Study of Biomolecular Binding in Polymeric Substrate and the Detection of Interaction among Biomolecules and Corresponding Binding Proteins and Conduction of the Diplomatic Assignment of "White Light Reflectance Spectroscopy for Detection of Biomolecular Interactions". She is Author and contributor numerous published papers in International peer-reviewed scientific journals, and responsible for oral presentations in National and International Conferences.

Christos Lagakis (male) is a highly experienced electrician having expertise in various areas such as CNC Laser robotics, world class industrial production, Total Quality Management, power circuits and Management of manufacture. His industrial experience ranges from Industrial automation, having worked at TITAN Cement LTD at this post, to construction and repair of PLC’s and CNC’s. Furthermore he holds expertise of the above focusing in the Naval industry.

Partner 5 - THERACELL	**THERACELL ADVANCED BIOTECHNOLOGY** www.theracell.eu
Organisation Description	

Theracell Advanced Biotechnology Ltd, is a UK biotechnology company, with a subsidiary in Greece, where its laboratories are located.

The company was established in 2011 with the aim to develop and provide advanced biotechnological applications, services and products in the field of regenerative biotechnology. In summary its areas of authority lies on:

1. Laboratory services in cell and tissue processing
2. R&D of new products & services leveraging on expertise in the fields of Molecular and Cellular Biology, Genetics, Biochemistry and Regenerative Medicine
3. Commercialization of biotechnological products, autologous or allogenic biological samples as well as pharmaceutical products and medical devices
4. Design and implementation of clinical trials.

Theracell laboratory is certified by ISO and GMP (Good Manufacturing Practice).

Company's mission is to provide innovative, safe, efficient, effective and affordable regenerative biotechnology products and cell therapies to medical community and consequently, patients. Company's vision is to become one of the leading and most influential regenerative biotechnology companies in Europe.

Theracell constantly exploits opportunities in regenerative biotechnology, using autologous adult stem cells and other cellular products. In order to develop and exploit the benefits of this exciting field, Theracell works closely with health professional experts and major medical and University institutions in EU countries. Theracell has vast experience in the isolation, processing and application of Adipose Derived Stem Cells (ADSCs), as well as somatic cells such as chondrocytes, fibroblasts etc. It has also developed several products based on the use of autologous growth factors that can be used in regenerative medicine and tissue repair.

With aim to "help people live a healthier life" the company operates with the highest standards of quality and technological advancement. Its people, well experienced scientists, share the same passion for true innovation through scientific excellence and integrity.

Relevant products

Current portfolio	Brand Name	Medical Specialty	Indication
Cell Therapies	Chodroseal	Orthopaedics	Sports Injuries/Chondral defects
	Cartil- S	Orthopaedics	Osteoarthritis/degenerative Chondral defects Injectable Autologous Activated Stem Cells
	Fibrolift	Dermatology/Plastic Surgery	Facial contour deformities
CytoFillers	Stemfill	Dermatology/Plastic Surgery	Facial contour deformities/Scars

R&D & clinical program:
Theracell constantly works on innovative approaches for the restoration of the lost cellular function of tissues and organs, in order to successfully address unmet medical needs. An ambitious R&D program is currently running in collaboration with research departments of Universities and specialized medical clinics. Theracell has at preclinical stage novel therapies in the fields of nephrology and neurology. Design and analysis of clinical studies required in each particular project falls under Theracell's responsibility and expertise.

Infrastructure/Equipment

Theracell currently employs 12 people mainly related with the cell production process and the R&D. The functions of Medical & Regulatory, Legal, Logistics and Marketing & Sales is outsourced from Qualia Pharma, which is a pharmaceutical company belonging to the president and major shareholder of Theracell.
Theracell is equipped with the latest automated technology and operates using a custom sample and data management software (**Datacore**).
It has established an Advanced Quality System which comprises the following **Accreditations and Certifications**:

- ISO 9001:2008
  Theracell's operations are certified by an ISO 9001:2008 by Lloyd's Register in the field of ***"Cell and Tissue Processing for Regenerative Medicine Applications".***
  The company's laboratories operate under the strict guidelines dictated by the ISO 9001:2008. These standards are based on a number of internal quality management principles ensuring customer (physicians & patients) orientation and satisfaction by providing consistent, highly quality products and services.
  Theracell is audited on an annual basis for maintaining its Advanced Quality System at the highest possible level.
- WHO-GMP
  Theracell was successfully audited and has obtained a **WHO-GMP (Good Manufacturing Practices)** certification in the field of "***Cell and Tissue Processing for Regenerative Medicine Applications***".
  It is among the very few biotechnology companies in EU with a GMP certification in the processing and production of advanced cell therapy products confirming its commitment to excellence.
  Good Manufacturing Practice (GMP) is a system for ensuring that products are consistently produced and controlled according to the highest quality standards.
  GMP covers all aspects of production; from the starting materials, premises and equipment to the training and personal hygiene of staff. There are detailed, written procedures in place that are essential for each process that could affect the quality of the finished product. These systems provide documented proof that correct procedures are consistently followed at each step in the manufacturing process – every time a product is made. WHO has established detailed guidelines for good manufacturing practice and many countries have formulated their own requirements for GMP based on these guidelines.
  - Clean Rooms Certified Operation

  The clean rooms of Theracell operate in accordance with ***ISO 14644-1*** standard and EU GMP Medicinal Products Annex 1. They are specifically designed for the advanced processing of cells and tissues for regenerative medicine applications.
  The classification of Thera state-of-the-art clean rooms is as follows:
  - Classification according to ISO 14644-1: Class 5
  - Classification according to EU GMP Annex 1: Grade B

  This certification exceeds the guidelines that govern the operation of Clean Rooms that are designed for bone marrow transplantation.

The clean rooms are constantly monitored to operate according to the above standards and are certified on an annual basis.

Profile of Main Persons

Fotis Sakellaridis, M.B.A. – President / Founder

Fotis Sakellaridis is a Chemist and holds an MBA from the MCE in Brussels and the IMD of Lausanne in the areas of Marketing and Management. Fotis is the owner and CEO of Qualia Pharma S.A. from 2009 till today. He started his career in 1988 as a Medical Rep of Glaxo S.A. From 1990 till 1997 he was employed by Astra Hellas S.A. where he held the positions of Product Manager, Marketing Manager and Marketing and Sales Director.

From 1997 till 2007 he served as Managing Director of Janssen-Cilag Greece S.A. In parallel, during the period 2002-2004 he served as Vice President Market Development Ortho Biotech Europe, M. East and Africa. He was the founding partner and CEO of Proton Pharma S.A. from 2008 till 2014. Fotis held various positions in many Industry associations, amongst others, Vice President of the Hellenic Association of Pharmaceutical Industry (SFEE), President of the Pharmaceutical Committee of the American-Hellenic Chamber of Commerce, Vice President of the Greek PhRMA, member of the Board of the Panhellenic Union of Pharmaceutical Industry (PEF), member of the General Council of the Hellenic Federation of Enterprises (SEV), member of the Greek Network of the Corporate Social Responsibility, member of the Association of Chief Executive Officers and President of the Greek Pharmaceutical Marketing Association (EEFAM).

Georgia Anastasopoulou - Managing Director

Georgia Anastasopoulou studied Business Administration and Marketing Management. She started her career in the pharmaceutical industry in 1990 passing thereafter, from all key Marketing & Sales positions of Medical Rep, Product Manager, Franchise Manager, Division Director, Company Marketing Director, BU Director, and General Manager that led her to the Managing Director position of today. During her career at the employment of Roche Hellas and Janssen – Cilag Greece she gained experience in the fields of OTC, Dermatology, Neurology, Psychiatry, Gastroenterology, Haematology and Oncology whereas, she had international involvement in specialized European Commercial Leading Teams in CNS & Biotech. In 2007 she assisted in the foundation of a new company, Proton Pharma, at the role of General Manager. Georgia is a member in a number of Associations such as Panhellenic Union of Pharmaceutical Companies, Hellenic Association of Marketing and Hellenic Association of Management.

Christina Ligoudistianou, Ph.D. – Scientific Director

Christina Ligoudistianou studied Biology at the Aristotle University of Thessaloniki. She holds a Ph.D in Immunology & Molecular biology from “Alexander Fleming” Biomedical Sciences Research Center. In 2003 she carried out part of her research work in the Laboratory of Molecular Enzymology at the Institute for Structural Biology “Jean-Pierre Ebel”, in Grenoble France, during an annual Marie Curie” scholarship.In 2004 she studied the effect of chemical compounds on human cancer cell lines, in a biotechnology company. In 2005, she worked in the laboratory of Dr. George Panayotou, Researcher of Molecular Oncology at the “Alexander Fleming” Biomedical Sciences Research Center. While studying the effects of chemicals compounds in human glioma and bladder cancer cells, she was also performing HPLC analysis in patients’ blood, for the pharmaceutical company WYETH. From 2006 till 2013, she’ s been working in ProCell SA, as a head of the laboratory for the isolation and preservation of umbilical cord blood and dental stem cells.

Christos Kiamouris, M.D., Ph.D. – Medical Director

Christos Kiamouris completed his studies at the Medical School of Athens and at the National School of Public Health. He specialized in Internal Medicine, in Hematology and in Oncology. At the same time he was trained in Methodology of Clinical Research, in Epidemiology and Statistics and on the Environmental and Occupational Medicine. Christos completed successfully a graduate program at Harvard University about Negotiating Skills and Telephone Medicine Methodology. He worked for the Oncological Department of “St. Anargyri” and Alexandra University Hospital Department. Christos has participated as a speaker at several international conferences and has worked with international centers in preparing Clinical Studies, focusing in neoplastic diseases. He has published his academic activities in many international magazines in the fields of Oncology – Hematology and Epidemiology. Since 2002 he works in the field of pharmaceutical companies as Medical Director both in Greece and abroad and Director of Clinical Research (companies Novartis, Janssen Cilag, Proton Pharma). Since 2013 he is the Medical Director of Theracell.

Maria Kontou - Laboratory Director Maria Kontou holds a Biochemistry degree from the University of Ioannina, Greece. She started her carrier at the National Center for Scientific Research (NCSR) “Democritos” as a collaborating scientist participating in a research program regarding growth and apoptosis in human glomerular epithelial cells. She continued her education at the Medical School of Athens for a master degree on “Clinical and Laboratory Medicine” and elaborated her project at the Genetic Research Center “Choremeio” at the Children’s Hospital “Agia Sophia” under the supervision of professor Dr. E. Kanavakis. Her thesis was involved with new technics for detecting carriers with Spinal Muscular Atrophy (SMA). After her graduation she worked from 2006 to 2013 in a private genetic center where she became an expert in cytogenetic analyses and cell cultures but also had the chance to be trained in all the departments of the center such as the Biochemistry and the Molecular Genetic department. From 2015 she’s been working in Theracell as a Laboratory Director.

Partner 6 - FU SAV	**Institute of Physics, Slovak Academy of Sciences Department of Metal Physics** www.fu.sav.sk
Organisation Description	

Institute of Physics of the Slovak Academy of Sciences (FU SAV) was founded in 1953 as Cabinet of Physics SAS and transformed into the Institute of Physics SAS in 1963. The main task of the Institute is to carry out basic theoretical and experimental research in physics. Research areas are solid state physics, quantum optics, nuclear and subnuclear physics. In solid state physics the effort is focused on rapidly solidified and metastable materials, (e.g. metallic glasses), and on special diagnostic and structure-analysis methods, a.o. scanning and transmission electron microscopy and X-ray diffraction, physical property measurements, atomic absorption and emission spectroscopy as well as computer materials science of metastable alloys. Current staff counts over 90 researchers (PhD. and DSc.) and 20 PhD students. The Institute is involved in a broad scientific collaboration within European scientific projects and cooperates closely with world most famous research centers (e.g. JINR Dubna, GSI Darmstadt, ESRF Grenoble, DESY Hamburg, CERN Geneve, ICTP Trieste).

Research activities of Department of Metal Physics (DMP) include kinetics, thermodynamics, stability, microstructure and catalytic properties of metastable alloy phases, also by ab-initio calculations and Monte-Carlo modelling, magnetization processes and magnetoelastic effects in soft and hard magnetic materials and applications of developed alloy systems. DMP owns diverse alloy melt processing furnaces and four planar flow casting devices and technology to prepare rapidly quenched alloys on laboratory and semi-industrial scale, also in protective atmospheres or vacuum.

DMP staff with 18 senior researchers, 5 postdocs, 5 technicians and 3 PhD students output in 2011-2015 over 200 CC publications and numerous conference contributions, also as a result of intense cooperation with partners from abroad, together with joint EU and NATO research projects and (co-)organization of 6 major international conferences.

DMP is co-founder of 3 centers of excellence SAS and also co-founder of C-MAC (European Integrated Center for the Development of New Metallic Alloys and Compounds) consortium. Cooperation with national research groups includes e.g. Institute of Experimental Physics SAS, Institute of Materials SAS, Institute of Materials and Machine Mechanics SAS, Slovak Technical University (Faculty of Materials Science and Technology, Faculty of Electrical Engineering), EVPU Nova Dubnica and a number of other small and medium enterprises. The number of national and international projects solved in the period 2010-2015 exceeds 35, including VEGA, APVV, bilateral and multilateral projects, 2 MNT-EraNet projects, FP6, FP7 and H2020 FET projects. Numerous cooperations with national industrial and applications sectors have also been realized, focusing a.o. on applications of amorphous and nanocrystalline alloys in power electronics, sensors, actuators and lightweight high-strength construction materials. Three patent applications from DMP have been submitted in this period related to preparation and physical processing of special rapidly quenched alloys.

Relevant Publications

- Škorvánek, J. Marcin, J. Turčanová, J. Kováč, P. Švec, Improvement of soft magnetic properties in Fe38Co38Mo8B15Cu amorphous and nanocrystalline alloys by heat treatment in external magnetic field. Journal of Alloys and Compounds 504 (2010) S135–S138.
- P. Šebo, P. Švec, D. Janičkovič, E. Illeková, Yu. Plevachuk, Interface between Sn-Sb-Cu solder and copper substrate, Mater. Sci. Eng. A 528 (2011) 5955–5960. E. Illeková, J. Harnušková, R. Florek, F. Simančík, I. Maťko, P. Švec, Peculiarities of the decomposition of TiH2. J. Therm Anal Calorim 105 (2011) 583–590.
- J. Hosko, I. Janotova, P. Svec, I. Matko, D. Janickovic, P. Svec Sr., Structure analysis of CoFeBSiNb(Ga) pseudobulk metallic glasses. Applied Surface Science 269 (2013) 77– 80.
- P. Svec, P. Svec Sr., J. Hosko, D. Janickovic, Formation of monophase Fe23B6-type alloy via crystallization of amorphous Fe-Ni-Nb-B system. Journal of Alloys and Compounds 590 (2014) 87–91.

Relevant Projects

- ULTRALIGHT – Application of advanced metallic materials for stiffness enhancement of lightweight structures (APVV - Slovak Agency for the Support of Science and Technology, 05/2011 – 10/2014, coordinator)
- NANOMORPH - Nanocrystalline and quasicrystalline metallic systems with tailored structure and morphology (APVV - Slovak Agency for the Support of Science and Technology, 07/2012 – 12/2015, coordinator)
- 3. HIGTEMAL, Bulk nanostructured Al profiles for applications at elevated temperatures (MNT ERA-Net Project, 01/2008-12/2010, partner, PI for IP SAS)
- NEW-MC-FROM-NCM - Progressive nanocrystalline and amorphous materials for application in selected high-power electronic devices (APVV - Slovak Agency for the Support of Science and Technology, 10/2013 – 07/2016, partner)
- FX-GATES - Physically processed rapidly quenched alloys for detection of low magnetic fields (SAS – TUBITAK bilateral project, 11/2013 – 20/2016, coordinator)
- NEXMAG- New Exchange-Coupled Manganese-Based Magnetic Materials (M-ERA.NET Transnational Call 2014, 10/2015-09/2018, partner)
- HELENIC-REF, FET-OPEN REA Project (06/2015-05/2018, partner)

Infrastructure/Equipment

- Transmission electron microscopes TEM JEOL 1200EX, JEOL 2000FX, JEOL 2011 equipped with image and diffraction analysis software, FEI Titan Themis with probe corrector and capability of chemical resolution of structures on atomic scales, attachments for in-situ heating experiments, precession electron diffraction for determination of high-resolution elemental and orientation maps, tomography. Complete line of TEM samples preparation devices.
- Databases and selected state-of-art software (even of own production) for image analysis, determination of chemical correlation maps of samples and electron diffraction analysis. Software for processing and modeling of atomically resolved structures.
- X-ray diffractometers Bruker D8 Advance with linear and point detector, Bruker-Siemens D5000HR with Euler cradle and parallel beam X-ray optics and Bartels and graphite monochromators, BRUKER GADDS, cold, humidity and hot sample chambers of own design to 600 and 1200°C.
- X-ray and electron diffraction analysis software – phase analysis with databases, morphology and quantitative analysis, microstress and texture analysis, Rietveld, structure refinement.
- Two laboratories for complete kinetic and thermodynamic analysis of materials (low and high temperature calorimetry) – DSC, DTA, thermogravimetry, calorimetry DSC with time-temperature profile modulation – MDSC, dynamic mechanical analysis – DMA for determining of complex mechanical parameters of solid substances: Calorimeters Perkin-Elmer DSC-7, DSC-8500, DTA-7 (with temperature range -40 to 1560°C) containing high-sensitivity furnaces, simultaneous dynamic temperature analyzer combined with thermogravimetric SDT Q600 and dynamic mechanical analyzer Q800 (TA Instruments), several magnetic thermogravimeters and specialized measuring and evaluation software.
- Numerous physical property measurement and annealing devices with wide range of time and temperature regimes for microscopic and macroscopic samples, magnetic field (up to 0.5T) annealing in protective atmosphere and vacuum.
- Device for measuring of density of low-volume (0.1-0.01 cm3) samples

Profile of Main Persons

Peter Svec Sr. (male), Dipl. Eng., Dr., D.Sc, senior scientist, head of DMP, vice-director of IP SAS, works at the Institute of Physics SAS since 1979. His main scientific interests are mass transport processes during phase transformations of metastable amorphous and nanocrystalline systems and quantitative transformation kinetics analysis and structure analysis using electron microscopy XRD techniques. He is author of three patents and two patent applications related to preparation of rapidly quenched multilayered ribbons, hot-shaping of metallic glasses and metallic glass sensors. He is author of 250+ CC papers, and about 90 other non-CC papers and presented over 90 invited lectures at international conferences. His work was cited more than 950 times according to the ISI database. He supervised 8 PhD students and has lead 6 diploma theses. He was awarded the SAS Prizes in 2012 and 2015 for international cooperation and progress achieved in the field of metal physics and the Prize of Jan Bahyl for successful implementation of Slovak patent for amorphous strain sensor. He was leader, deputy leader or project member of numerous (50+) national, bilateral and multilateral international projects funded by Slovak and foreign grant agencies, MNT ERA.Net and M-ERA.NET, H2020 (FET), Centers of Excellence SAS, ASFEU-Structural Funds, etc.

Peter Svec (male), Dipl. Eng., PhD, post-doc under 35 years, Ph.D in solid state physics (phase transformations in amorphous systems based on Fe and Ni, 08/2012), since 01/2013 holder of SAS post-doc. fellowship. Implementation of trivariate correlation analysis to chemical mapping in SEM and TEM, development and adaptation of image processing systems for TEM and of precession electron diffraction system for nanoscale phase mapping in TEM. Post-doc stay at SuperSTEM Laboratory Daresbury (2013), skills in installation and operation of atomically resolved STEMs and image/spectroscopy datacube processing and multivariate correlation analysis, development of site-specific sample preparation for STEM observations. Skills in preparation of rapidly quenched samples, development of method of embrittlement-free hot shaping of metallic glasses. Skilled experimentalist, expert in X-ray and electron diffraction and structure analyses, author of 30+ CC publications, leader of national projects and of bilateral cooperation project with SuperSTEM Laboratory.

Irena Janotova (female), Dipl. Eng., Ph.D, early stage researcher, post-doc under 35 years, Ph.D in physical engineering, author 20+ CC publication and numerous other refereed contributions, expert for design and preparation of soft magnetic metallic systems and optimalization of their thermomagnetic properties, expertise in thermodynamic, thermomagnetic and kinetic characterization of phase transitions from metastable (amorphous) state.

Juraj Zigo (male), Mgr. in solid state physics, PhD student (supervisor Dr. Peter Svec Sr.) in solid state physics, 09/2013 – 08/2017, focusing on local ordering on atomic scales of metastable alloy systems and interactions with nanoparticles, adsorbent atoms and molecules.

Marian Krajci (male), Dr., DSc., graduated with honors from Comenius University in solid state physics, PhD at FU SAV in 1987, DSc. at Comenius University in 2005. Numerous long and short-term stays in Japan (Prof. T. Fujiwara) and Austria (Prof. J. Hafner) related to ab-initio determination of quasicrystal structure and catalytic properties of complex metallic alloys. He has contributed to explanation of the character of quasicrystal electron spectrum (1992). He participated in building-up of model of Al-Pd-Mn quasicrystal (1995), in explaining semiconductor behavior of several aluminium – transition metal alloys (2007). He focuses on studies of physical, chemical and mainly catalytical properties of surfaces of complex metallic alloys and their nanoparticles (together with prof. A. P. Tsai, Sendai, Japan). He authored over 95 CC publications with over 1000 WOS citations (hindex=20). He was awarded the Slovak Academy of Sciences Prize for international cooperation.

Partner 7 - PNO	
	PNO Innovation NV www.pnoconsultants.com

Organisation Description

General Introduction to the PNO Consultants group of firms

PNO is Europe's largest independent public funding and innovation consultancy with 30 years of hands-on expertise with more than 500 funding programmes in most EU countries, annually raising approximately 1 Billion Euro for its clients. Created in 1985, PNO is a high-growth knowledge-intensive company, supporting over 2000 clients throughout Europe, annually developing over 250 European consortia. PNO has presence in 6 European countries, employing in around 250 people in western Europe alone.

PNO's "Innovation Management" services delivers high quality support to large sized companies, SMEs, Universities, Research Institutes, Associations and clusters in the full cycle of the innovation process, including:

- Analysing, defining and planning innovation processes;
- Building innovation networks, partnerships and projects;
- Managing Projects and driving innovation.

PNO within the BioEconomy Space

PNO has a strong client base and track record in the interlinked chemical industry, biotech and eco-innovation sectors, with a particular insight in EU policies and programmes, as well as innovation and funding projects and programmes in 6 European countries. Being partner in multiple EU-funded (including FP7 and Horizon 2020) projects and several commercial assignments, PNO Consultants fully qualifies as a relevant and positive long-term stakeholder within the European BioEconomy panorama. For example, PNO is active partner in the FP7 bio-projects BIO-TIC and BIO-MIMETIC as well as in the Horizon 2020 bio-projects BIOLINX and ROBOX. On a commercial base, PNO has recently produced a practical guide to support the Biobased Industry crossing bridges and optimizing funding possibilities. To help this sector make maximum use of the synergy possibilities, the Bio-based Industries Consortium (BIC) and PNO Consultants (PNO) published the practical guide "Combining BBI (H2020) and European Structural and Investment Funds (ESIF) to deploy the European bioeconomy". In addition to that, PNO is involved as advisor in roadmap development for the chemical industry towards Public Private Partnership Programmes in HORIZON 2020 (eg: SPIRE), with an important focus on industrial biotech, as well as European networking and incubator projects with various smart specialisation regions which will provide a real life experience base. Finally, PNO has supported the European Commission DG GROWTH Expert Group on BioBased Products on the evaluation of a continental scale Biobased Product Public Procurement Program.

Relevant Publications

1) Analysing, defining and planning innovation processes

- Market, stakeholder & Technology Intelligence. Provides a helicopter view on who's doing what in the EU and beyond, benchmarking target technology with other technologies and solutions, analysing and identifying the essential markets that to be targeted;
- Systematic Innovation using advanced methodologies (eg: TRIZ). Discovers new technologies, partners and markets, structuring and driving open innovation processes in multi-stakeholder contexts, or for individual clients;
- Vision and roadmap development. Shows where to focus, organising, supporting and reporting on vision and

innovation roadmaps of interest.

2) Building innovation networks, partnerships and projects

- Develop innovation projects, proposals and consortia. Builds innovation project with international partners supported by public funding, bringing the best funding expertise, relevant networks and hands-on support;
- Effective linking to innovation ideas, partners and funding across sectors and Europe. Enables becoming a partner in projects supported by public funding to discover or bring innovation and technology to new markets, unlocking networks of projects and stakeholders.

3) Managing Projects and driving innovation

- Drive the research project by market and users' requirements. Determines who could be interested in the generated results and findings, helping discovering with structured analyses and reporting;
- Technology valorisation and transfer to the market with effective and efficient exploitation and dissemination strategies. Enhances go-to-market approach, acting as business developers, supporting technology towards new markets.

Relevant Projects

PNO has a long-lasting experience in successfully developing European projects including FP7, LIFE and ecoinnovation projects bridging the gap between research and market development. PNO has been the architect and is currently participating in a number of industry Support networks with Support from the European Commission in the Security, Transport, Biotech and Chemical sectors, promoting the establishment of industry collaborations, knowledge transfer and technology acceleration. **Within the BioEconomy space these include:**

1) **BIO-TIC** (FP7) - Roadmapping for Industrial Biotech to overcome hurdles towards 2030 DG Research Biotech companies 2012 to 2015 (1,9 MEUR);
2) **R4R** (FP7) - Technology Innovation System analyses and development of resource efficiency projects in 6 chemical industry regions in Europe DG research Chemical industry, biotech industry 2012 to 2015 2,5 MEUR;
3) **SPLASH** (FP7) - Stakeholders analysis for a new biobased industrial platform using microalgae as a renewable raw material for the sustainable production and recovery of hydrocarbons and (exo)polysaccharides 2012 to 2016 12.1 MEUR
4) **ROBOX** (H2020) – Stakeholder analysis for improved techno-economic viability of bio-transformations via four types of robust oxidative enzymes 2015 to 2019 11.3 MEUR;
5) **BioLinX** (H2020) – Stimulate Innovation, networking and brokerage by initiating range of powerful linking and innovation acceleration processes 2015 to 2018 2.0 MEUR.

Infrastructure/Equipment

PNO will made available to the project its virtual Open Innovation Platform www.innovationplace.eu, proprietary on-line networking, community building, management and dissemination tool, hosting more than 15.000 registered users, all with a relevant innovation profile.

Profile of Main Persons
PNO Innovation - Management 1) **Ron WEERDMEESTER**, Male, Lawyer and Certificate in International Business Programme, graduated from Rijksuniversiteit Groningen. He also holds a Certificate Degree in "Information Systems in Organisations", from Boston University. He is Business Manager for EU grants and responsible for Development of strategic (technology development) projects with a European dimension, development of R&D projects together with industry and universities, project management, RTD projects in FP4,5,6 and 7. In detail, he has gained substantial experience as a specialist in EU RTD and innovation projects through 15 years of innovation and technology transfer consultancy. He has been involved in the development of more than 40 successful EU projects, including LIFE and eco-innovation. He has specific experience as a project manager of various European Commission funded Economic and Technological Intelligence actions focused at assisting European industry on their innovation needs and developing European networks to realize their innovation strategies through internationalization and EU projects. He has been responsible for an overall promotion of more than 300 industry technical audits and the promotion of more than 80 European projects. As an expert in European network development, he regularly gives training to SMEs coming from various European countries, on issues as European project management in networked organisations, technology transfer and the development of European partnerships. PROFILE: https://www.linkedin.com/in/ronweerdmeester **PNO Innovation - BioEconomy Expertise** 2) **Filippo Giancarlo MARTINELLI** (PDEng, MSc), Male, currently works as Grants and Innovation Consultant at PNO Innovation, Brussels. He is mostly involved in proposal writing, project participation and business development in the field of BioEconomy (eg: BBI) and sustainable process industry (eg: SPIRE), including a role in roadmap integration for the BIO-TIC FP7 project and support to the EC DG GROWTH BioBased Products Expert Group. His prior experiences include: Product Manager at Agroils Technologies (biodiesels), Florence, from Feb 2014, BioProduct Designer at Delft Advanced Biorenewables (biofuels), DSM (biochemicals) and TU Delft, Delft, from Feb 2012 and Research trainee at KtedoGEN/NAICONS (bioactives), Milan, from July 2010. His education includes a PDEng (industrial PhD) from TU Delft, an MSc from University of Milano-Bicocca, an International Summer School from ETH/UZH, Zurich, and a BSc from University of Parma. He is Board Member at the National Association of Biotechnologists (ANBI, Italy) as well as Work Group Member (WG4 Policy and Sustainability) at the European BioFuels Technology Platform (EBTP, Brussels). Recently he co-founded the online database BioDirecta: “All Biotech Companies at once!”. PROFILE: https://www.linkedin.com/in/fgmartinelli

Partner 8 - IRIS	
	INNOVACIÓ I RECERCA INDUSTRIAL I SOSTENIBLE SL www.iris.cat
Organisation Description	
IRIS was established in Spain in 2007 as an advanced engineering and R&D company, specialising in process engineering. IRIS specialises in three fields of engineering: Production Processes (with strong specialisation in novel technologies); Smart Systems; Optical Systems, combined with knowledge and research capacity in food science, materials science and environmental science. IRIS applies its knowledge and expertise to develop solutions for industry in the fields of advanced monitoring and control, surface activation and decontamination, as well as waste valorisation strategies. IRIS also provides contract research and advanced engineering services to the food, agriculture, chemical and pharmaceutical industries, among others. The company has a policy of continuous investment for growth, and currently employs 48 staff based in two sites (Barcelona and Dublin). IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i.	
Relevant Publications	

Publications in relation to Innovation Management:

- O. Mc Nerney, A. Stachowicz, P. Czuprynski, E. Philpott, D. Kolman, D. Martin, E. Herrmann (2010). Strategic Guide to Successful Use and Dissemination of the Results of Research and Development Projects. 1st edition, Steinbeis-Edition Stuttgart 2010. ISBN 978-3-941417-27-4

Publications related to valorisation of waste streams:

- O. Mc Nerney, A. Gimenez (2011). Proyecto Geronimo II - Biogás: Transformando los residuos de las granjas en energía. Revista RETEMA. March/April 2011. Vol. 149
- "Processing and Validation of Whey-Protein-Coated Films and Laminates at Semi-Industrial Scale as Novel Recyclable Food Packaging Materials with Excellent Barrier Properties"; E. Bugnicourt, M. Schmid, O. Mc. Nerney, J. Wildner, L. Smykala, A. Lazzeri, P. Cinelli.; Advances in Materials Science and Engineering, vol. 2013, Article ID 496207, 10 pages, 2013.
- "Properties of whey protein coated films and laminates as novel recyclable food packaging materials with excellent barrier properties", M. Schmid, K. Dallmann, E. Bugnicourt, D. Cordoni, F. Wild, A. Lazzeri, K. Noller, International Journal of Polymer Science, vol. 2012, Article ID 562381, 2012.
- "Maxi-use of wastes from agro-food processing to obtain truly sustainable bioplastics", E. Bugnicourt, O. Mc Nerney, A. Lazzeri, Bioplastics magazine, Vol 6,

Relevant Projects

- Project acronym: WASTE2FUELS; Project full title: Sustainable production of next generation biofuels from waste streams. Grant Agreement no. 654623. January 2016 – December 2018. Funding programme: H2020. Duration: 36 months.
- Project acronym: MARE-Valorisation of hydrocarbon sludges for the production of additives for marine remediation and biomass for energetic purposes/MANUNET-2011 R21104/Jan 2012-July 2014
- Project Acronym: OLI-PHA. Project full name: A novel and efficient method for the production of polyhydroxyalkanoate polymer-based packaging from olive oil waste water. Grant Agreement no: 280604-2; Duration: 01.06.2012-31.05.2015. Funding programme: FP7
- Project Acronym: PROPAT. Project Full Title: Robust and affordable process control technologies for improving standards and optimising industrial operations. Grant Agreement no: 637232 (in grant preparation). Funding Programme: H2020. Duration: 48 months.
- Project acronym: WHEYLAYER; Project full title: Whey protein-coated plastic films to replace expensive polymers and increase recyclability; Grant agreement no.: 218340-2; Nov. 2008-Oct. 2011. FP7 (we developed the coating line and in the subsequent WHEYLAYER-DA project we performed the industrialisation of the line).

Infrastructure/Equipment

Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others. Food and water testing and wet chemistry laboratory equipped with HPLC with UV-Vis detector, DSC machine, NIR-VIS spectrophotometer, climatic chamber and microbiological safe area among others. In terms of programming and advanced ITs capabilities, IRIS has a specialized staff (software and telecommunication engineers with programming experience) that work on the basis of the Agile methodology and philosophy (scrum and kanban) for developing complex applications that include hardware interfaces, SQL and No-SQL databases (MySQL, PostgresDB, Couchbase among others) and advanced and user-friendly graphic user interfaces not only for computers (using Windows Presentation Foundation) but also for smartphones (Android, iOS) or web-based access (HTML 5). In order to work under scrum and kanban methodologies, IRIS has dedicated project corners and sprint rooms in which programming teams can develop their duty in a creative atmosphere.

Profile of Main Persons

MR PAU PUIGDOLLERS DE BALLE (Male). BSc in Environmental Science from the Universitat Autònoma de Barcelona. MSc in Terrestrial Ecology (Research Thesis: "Effects of biochar from fast pyrolysis on soil water availability and basal respiration", 2010) from the same university and MSc in Renewable Energy Engineering and Management from the Universitat de Barcelona. Prior to joining IRIS he held positions as technical sales person on Servicios Normativos consultancy group, quality field inspector on waste treatment plants for Ingeniería IDOM International, performing quality audits and reports. He has also held different research positions at the CSIRO in Australia and in the CREAF in Barcelona. He has also experience as environmental consultant at Eco Intelligent Growth, working with renewable energies, life cycle assessment, C2C certifications and green building.

Dr. LAURA RODRÍGUEZ TURIENZO (Female). B.S in Food Science & Technology (2008) Masters in Innovation in Safety & Food Technologies (2010), PhD in Biological Sciences (2011), all from University of Santiago de Compostela (USC), Spain. She has participated in a number of R&D regionally and nationally funded projects aimed at developing and validating new bio-based active edible packaging, physico-chemical characterization of novel foods and valorisation of food industry wastes. She is author of several papers in peerreviewed journals and has participated in different national and international conferences.

MR JOAN A. MARTÍNEZ BORRÀS (Male). Telecommunications Engineer from the Universitat Politècnica de Catalunya specialising in Electronic Systems. Broad experience in the development of embedded systems working with microcontrollers from Intel, Nec, Motorola, National Semiconductor, Zilog, Renesas (Hitachi) and Atmel and its related software written in C and assembler. Expertise in R&D and industrial development- Indomatic Elite, Dam Alignment By Means Of Laser for Endesa and test devices (control of the electropneumatic test device for the SEAT Toledo dashboard manufactured by VDO). His communications libraries following the V.21, V22, V.22bis and X.25 standards for Electronic Funds Transfer terminals were deployed on a few million payment terminals around the world and his teleloading system allowed some hundreds of thousands of terminals to collect the transactions and update their software online. He also has led the development of some security EFT terminals following the EMV, ZKA, VISA PED and PCI PED standards. His programming skills include assembler, C, C++, Java, Visual Studio, .NET, Linux, Qt, Gtk, HTML, CSS, Javascript, PHP.

Dr. ANA GARCÍA SANZ (Female). PhD degree in Biophysics from the Autónoma University de Madrid (PhD thesis: Electron and positron interactions with molecules of biological interest"). Masters in Biophysics from the Autónoma University in Madrid and Master-degree in Physics from the Complutense University de Madrid. Prior to joining IRIS, she has worked at the National Research Council Centre (CSIC) and has been invited as a research intern in Universitá La Sapieza di Roma, The Open University and Sherbrooke University. Her research has been focused on the study of the damage induced by radiation (electrons, positrons and X-rays) in biological systems at molecular level, by means of both theoretical (quantum chemistry calculations) and experimental (spectroscopy, HPLC, LC-MS/MS) techniques. Author of 17 scientific papers in peer reviewed Journals (J. of Physical Chemistry Letters, Radiation Research, Physical Review A, Journal of Chemical Physics).

MR DAVID MONTES MONTESERIN (Male). BSc in Mechanical Engineering (2011), Polytechnic University of Catalonia (UPC), Barcelona. Prior to joining the Production Processes group at IRIS as Mechanical Design Engineer, he gained wide experience in product design and development, including dosing systems for the chemical industry and both industrial and domestic appliances for the water treatment sector. He has also held positions as Engineering Team Manager, applying renewable energies to industrial-sized drinking water plants among other projects. He is specialised in design for manufacturing and is proficient in SolidWorks, design software which has taught to several companies as a freelancer

Partner 9 - EXERGY	**Exergy Ltd.** **Energy & Renewable Engineering** **Sustainable Processes Department** www.exergy.uk.com
exergy	

Organisation Description
Exergy is one of the UK's leading renewable energy companies with the expertise to provide services, across the globe, to develop, engineer, construct and operate projects that contribute to our goal of a low carbon, sustainable future. We have been in the industry for over 10 years, contributing to the development of the global renewable energy market. We have helped to move the sustainable energy debate from the margin to the mainstream. Operating across a range of innovative low-carbon energy technologies, our apabilities cover everything from project inception to energy generation and operation. Sustainability lies at the core of our values; balancing commercial, environmental and social concerns to address the world's energy problems is what makes us different. We are pioneers of Renewable Energy and Green-Construction. We are unique in our ability to offer vertically integrated skills in a range of sustainable energy technologies. Exergy can be involved at every stage: from project design and development, through planning and financing, engineering and construction to operation. Exergy ensures schemes are designed and built to generate energy in the most economical way and with minimal environmental impact. Exergy provides a range of technical and professional services across the energy and built environment sector by means of a multi-disciplinary team supported by extensive practical experience in the following specialist areas: Feasibility Assessments that include techno-economical assessments; Design, Modelling and Scaling up of Chemical and Biochemical Processes; Resource Efficiency & Environmental Management; Water Management Strategies; Design and Delivery of Resource Efficiency support for SME's; Development, Monitoring and Measurement of Sustainability Projects; Low Carbon footprint Construction Techniques and Life Cycle Assessment. Exergy has a strong work team made up of different profiles: chemical and process engineers, mechanical engineers, civil engineers and environmental engineers, amongst other. The Sustainable Process Department utilises industry-leading software for the simulation, optimisation and upscaling of chemical processes enabling us to offer process engineering services focusing, but not limited on the following areas: Process design and operations improvement. Protein synthesis and purification process. Protein extraction process. Gasification, methanation, pyrolysis. Energy and emissions reduction. Debottlenecking. In biological processes, among others, Aspen Plus and SuperPro Designer can be used for simulating chemical and biochemical processes. The simulation results can be complemented with a group of key parameters to complete the scale up and industrialization of the processes. Aspen Plus can be used for simulating different stages of the production of biochemical and biofuels. We combine the best software with our process design knowledge to improve the profitability and efficiency of both new and conventional processes. Our experience allows us to identify opportunities for improvement and develop innovative and sustainable solutions across the process sector. We utilise various process simulation tools early in the design process which enables us to see the economic impact of design changes earlier in process development and identify opportunities for capital and operating cost savings so we can design the optimum process according to the project requirements. We have a wide array of tools at our disposal enabling us to construct and optimise process models and produce required engineering documents (specification, equipment lists, PFDs, preliminary P&IDs). Exergy is also full member of SPIRE PPP and BIC.
Relevant Publications

- Simulation of chemical pre-treatment process and Optimisation and scale up of the system based on simulation. Liquid and gas Fischer-Tropsch fuel production from olive industry waste: fuel from waste. Exergy Ltd, report for the European Commission. 2015.
- Report on third generation bio-fuels (2016). Results from third generation biofuels production including Microbial Fuel Cell (MLC) will be included in this deliverable. Exergy Ltd, report for the European Commission. 2016.

- Sustainable techno-economic solutions for the agricultural value chain. Ethanol production at demonstrative scale assessment. Results and discussion about the feasibility of the installation at demonstrative scale. Exergy Ltd, report for the European Commission. 2016.
- Scale-up and design of an optimised algae process for biodiesel and biogas production. Molto Marin, Jose L. Chemical Engineering, Thesis - Madrid, 2015.
- Report on sustainable production of next generation biofuels from waste streams. Risk assessments. report for the European Commission. Exergy Ltd, report for the European Commission. 2016.

Relevant Projects

- FFW (FP7 ENV, No. 308733) www.fuelfromwaste.eu, "Liquid and gas Fischer-Tropsch fuel production from olive industry waste: fuel from waste". The overall aim of the project is to use agricultural waste, mainly residues from olive farming and olive oil production, for the generation of ready-to-use fuels, namely synthetic natural gas (SNG) and diesel, which would be primarily used for energy production for oil manufacturing and as transportation fuel for olive farming.
- INNOBITE (FP7 ENV NO. 308465) www.innobite.eu, Transforming urban and agricultural residues into high performance biomaterials for green construction". The project will transform urban and agricultural residues into high performing, resource efficient products for the emerging Green Construction sector. The project is supported by two innovative ideas: Adding value to the inorganic fraction of wheat straw. Obtaining cellulose nanofibres (MFC) out of highly recycled paper.
- AgriMax (BBI JU – In negotiation stage) "Agri and food waste valorisation co-ops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications" Exergy lead the engineering, design and mass and energy balances for both pilot plants.
- WASTE2FUELS. (H2020 No 654623-2). "Sustainable production of next generation biofuels from waste streams". www.waste2fuels.eu. Exergy role: Risk assessment and IPR expert on Biofuels.
- AGROCYCLE. (H2020 No. 690142). "Sustainable techno-economic solutions for the agricultural value chain". EXERGY leads the WP Biofuel Production and leads the Pyroloysis of Forest Residues and Third Generation Biofuels. Exergy also contributes to the development of microbial fuel cells and to the relevant lab activities.

Infrastructure/Equipment

- AspenONE® Engineering Desktop/SuperPro Designer. Exergy utilises industry-leading software for the simulation and optimisation of chemical/biochemical/enzymatic processes.

Profile of Main Persons

Fernando Centeno, PhD Energies and Energy Efficiency (Male). Mr. Centeno has wide experience on the engineering and consultancy field related to the building construction & energy sector, such as novel multifunctional materials, renewable energy technologies (solar PV, geothermal and biomass) and energy efficient programs for buildings, districts and industrial facilities. He has been working for many years in R&D projects at national and European level on the Energy and NMP areas. Member of the "*Sustainable Process Industry through Resource and Energy Efficiency*" (SPIRE-PPP) association and European Construction Technology Platform (ECTP). Has been involved in two FP7 coordinations focused on the renewable energy sector. As Head of the R&D Group, he has participated in more than 38 international R&D project (i.e. H2020, FP7, IEE, Life+, Eco-innovation, Eureka, Eurostars, among others).

Rocio Roldan, MSc (Female): is Chemical Engineer by Complutense University of Madrid. Her expertise area is related to the modelling and simulation of chemical and biochemical processes and unitary operations, especially focused on the AspenOne software handling and mathematic modelling. She has developed and enhanced other strong skills along her professional experience together with further training: Petrochemical simulation and gas cleaning (ProMax®); Power Plant Operator; Circular Economy and Bio-Technology (Delft University), amongst other. She has been involved in FP7 projects related to the valorisation of biomass into valuable products and related to circular economy: FFW project (simulation and scale-up of the entire process; optimisation of the equipment and consumptions; development of case studies; and energy integration by means of *Aspen Energy Analyzer*), INNOBITE and CloseWEEE project. Besides, she has been in charge of technical writing and coordination in H2020 proposals.

Jose Luis Molto, MSc (Male): holds a master in Chemical Engineering by Rey Juan Carlos University. He has a strong background in simulation, optimisation and modelling of chemical and biochemical processes (Aspen HYSYS, Aspen PLUS and SuperPro) along with the feasibility and economical assessments of them, which has been strongly enhanced and extending during his professional career, especially focused on biochemical processes and complemented by independent courses and training: (ProMax® 3.2-Bryan Research & Engineering; Technology for Biobased Products - Delft University in Netherlands, among others). One of his main areas of specialisation is the algae biofuel (biogas and biodiesel) from photobioreactors and its scale-up and optimisation through SuperPro designer. He is an active member of the European Federation of Chemical Engineers (EFCE) since 2013. He has optimised and developed the simulations of the industrial scaled up process in the European FFW project and the different cooling methods and the use of residual heat recovered for a photon-enhanced thermionic emission process in FP7 ProMEThEUS project. Besides, he has participated in the proposal phase of H2020 projects which are currently in evaluation stage/developing for submission

Partner 10 - VERTECH	
Vertech Group	**VERTECH GROUP** www.vertech-group.com

Organisation Description

VERTECH GROUP (VERTECH) is a French research intensive SME with business focus on High-Tech Environmental Consultancy, International R&D Consultancy and Market Research and business development. VERTECH aims to create value by supporting and advising private and public organizations in various sectors (e.g. recycling, energy and waste treatment), and regions to get from research to market. VERTECH is also present in Spain and has expansion plans to other markets besides Europe (e.g South America).

VERTECH capabilities include:

- Sustainability Assessments of processes and strategies.
- Analysis of ecological footprint.
- Life Cycle Assessment and Life Cost Cycling of innovative processes and products Environmental Technology Verification.
- Socio-economic impacts of innovative products by Social Life Cycle Assessment.
- Techno-economic, environmental, social and legal analysis and risk assessment.
- Exploitation of R&D results and exploitation plans.
- Dissemination activities.

Vertech Group is member of the French Life Cycle Assessment Platform (Avnir). The company is based in Nice, in the French EcoValley, that is the first example of a real sustainable urban environment.

The staff of Vertech Group has over 10 years of experience on international cooperation, coordinating international R&D projects and WPs. Currently Vertech participates in 12 R&D FP7/H2020 project leading the environmental assessments and supporting in the exploitation of results. Similar role that in this project. Vertech does the technical leading of energy projects such as FFW project and the environmental assessment in the 12 project in which participates. The main role of Vertech will be leading the Life Cycle Thinking in WP5, from the environmental (Life Cycle Assessment -LCA), social (Social Life Cycle Assessment - SLCA) and economic (Life Cost Cycle - LCC) points of view. Vertech will also support in the technical coordination of the project given its expertise in FFW project.

Relevant Publications

- Garfí M., Cadena E., Pérez I. and Ferrer I. (2014). Technical, economic and environmental assessment household biogas digesters for rural communities. Renewable Energy, 62, 313–318.
- Adesanya V., Cadena E., Scott S. and Smith, A. (2014). Life cycle assessment on algae biodiesel production using a hybrid cultivation system. Bioresource Technology, 163, 343-355.
- Colón J., Cadena E., Sánchez A., Artola A. and Font X. (2012). Environmental impacts of installations of biological treatment of municipal organic wastes. Ingeniería Química, 44, 62–67. Colón J., Cadena E., Pognani M., Maulini C., Barrena R., Sánchez A., Font X. And Artola A. (2013). Toward the implementation of new regional biowaste management plans: Environmental assessment of different waste management scenarios in Catalonia. Resources, Conservation and Recycling. Volume 95, February 2015, Pages 143–155

Relevant Projects

- GRAIL - Glycerol Biorefinery Approach for the Production of High Quality Products of Industrial Value. FP7 KBBE (Ref.: 613667).
- FFW - Liquid and gas Fischer-Tropsch fuel production from olive industry waste: fuel from waste. FP7 Environment (Ref: 308733).
- NOSHAN - Sustainable Production of Functional and Safe Feed from Food Waste. FP7 KBBE (Ref.: 312140)
- INNOBITE - Transforming urban and agricultural residues into high performance biomaterials for green construction. FP7 ENVIRONMNET (Ref.: 308465).
- CloseWEEE - Integrated solutions for pre-processing electronic equipment, closing the loop of postconsumer high-grade plastics, and advanced recovery of critical raw materials antimony and graphite. H2020 WASTE (Ref.: 641747).
- AquaNES - Demonstrating synergies in combined natural and engineered processes for water treatment systems. H2020 -WATER (Ref.: 689450)
- NANOPACK – Pilot line production of functional polymer nanocomposites from natural halloysite nanotubes: demonstrating controlled release of active antimicrobials in food packaging applications. H2020-NMBP-PILOTS .
- KARMA2020 - Industrial Feather Waste Valorisation for Sustainable KeRatin based MAterials. H2020SPIRE.

Infrastructure/Equipment

Software SimaPro® 8 (PRé Consultants) and Ecoinvent v.3 databases (Swiss Center for Life-Cycle Inventories, 2010). Analytical tools in accordance with ISO 14040 & 14044 standards (ISO 14042, 2000).

Profile of Main Persons

Dr Jose Antonio Gutierrez-Bravo (Male)

Dr. Jose A. Gutierrez has a PhD in Environmental Geology and 5 MSc: MSc in Energy Resources, MSc in Renewable Energy, MSc in Mining Resources, MSc in Business Administration and MSc in Health, Safety and Environment. He has more than 8 years' experience in research activities on energy and environment issues and he has worked in 6 different R&D centres and companies, performing R&D, including: Spanish Institute of Geology and Mining, University of Leon, CIUDEN es.CO2-Tecnology Development Centre for CO2 Capture, Vertech, among others. He has participated in more than 15 European R&D Projects developing and validating clean energy and environmental solutions.

Dr Paola Castrillo (Female)

Paola Castrillo has a Science and Materials Engineering PhD from Carlos III University of Madrid. From 2004 to the present she has been active in the preparation, management, coordination and / or development of several European R & D projects in the areas of sustainability and revalorization of urban and agricultural wastes/residues (i.e. FFW-FP7, NOSHAN-FP7, GRAIL-FP7, CLOSEWEEE-H2020, INNOBITE-FP7, TECNOCAI- CENIT, etc.). Her research interests include sustainable revalorization of urban wastes, energy and resources efficiency, sustainability and manufacturing processes efficiency assessments. She is the Technical Coordinator of the FFW (FP7) and CLOSEWEEE (H2020) projects. Currently working as Commercial Manager for Vertech Group. Support on LCA, LCC and social LCA development.

Dr Erasmo Cadena (Male)

Erasmo Cadena has a PhD in Environmental Science from Universitat Autónoma de Barcelona (UAB). He holds a Msc in Enviromental Engineering. Postdoctoral Researcher at University of Cambridge-UK and Universitat Politècnica de Catalunya (UPC), BarcelonaTech. He has experience in international research initiatives and his research interests include technical, environmental and economic analysis of new processes and high-performance. products, life cycle assessment, life cycle costs analysis. Erasmo will be performing the LCA tasks in WP8.

Eléonore Chabbey (Female) Eléonore Chabbey is BSc in Business & Management from the University of Nice Sophia Antipolis, Msc1 and

Msc2 in Economy Management of Small-Medium-Enterprises. Currently Mrs. Chabbey is in charge of business development within the company, her activities include market research and analysis, business modelling and market uptake strategies. **Antonio Barona (Male)**

Antonio Barona is the responsible of the overall R&D, Innovation & Business strategy of the company. He holds a Bs in Civil Engineering, M.Sc. on Materials and Innovation, M.Sc. in Project Management. Mr. Barona has more than 10 years of experience in the coordination of international research. He is member of several international academic, business and research platforms dealing with Environmental and Energy Technologies for a Sustainable Future. In terms of international research cooperation, he has not only actively participated in several project proposals but has led the project Coordination of the FP7 projects: EE-QUARRY & S4EeB. He will be performing the management of the project and WP7, assessing on strategic planning and support on LCA and LCC.

Partner 11 - EPARELLA	
	eparella-GmbH **(**limited liability company**)** www.ecoduna.com

Organisation Description

Eparella GmbH, 100% daughter of the ecoduna AG, is a young SME biotechnology company which has developed and patented a procedure for the production of microalgae at industrial scale and now makes use of this proprietary technology to attain the market leadership as a producer of microalgae products.

Since 2010 ecoduna has developed and tested a novel photobioreactor technology (hanging gardens) and has lots of experience in the 'mid to large sale cultivation' of microalgae, as well as photobioreactor design. The team of 22 people initially working at ecoduna produktions GmbH now are employed at eparella GmbH. The team with its long lasting expertise consists of engineers, process technicians and biologists and molecular biologists developing and operating photobioreactors (PBR) for microalgae cultivation.

ecoduna has spent the last 6 years developing a unique, world-patented sustainable production process for the efficient and sustainable generation of high quality microalgae. Only the integration of unique technological expertise combined with biological expertise and data management enables ecoduna/eparella to offer consistently high-quality biological resources in large quantities for the processing industry.

Relevant Publications

Kopecky, D., Weichselbaum, O., Fluch S., (2012) MAPS – A service-oriented, customizable, multi-purpose LIMS; Nature Methods | Application Notes June 2012; ISSN: 1548-7091

Bräutigam K, Vining KJ, Lafon-Placette C, Fossdal CG, Mirouze M, Marcos JG, Fluch S, Fraga MF, Guevara MÁ, Abarca D, Johnsen O, Maury S, Strauss SH, Campbell MM, Rohde A, Díaz-Sala C, Cervera MT; (2013) Epigenetic regulation of adaptive responses of forest tree species to the environment.; Ecol Evol. 2013 Feb;3(2):399-415

Elisabeth Wischnitzki, Eva Maria Sehr, Karin Hansel-Hohl, Maria Berenyi, Kornel Burg and Silvia Fluch; "How to isolate a plant's hypomethylome in one shot, Biomed Research International, Volume 2015 (2015), Article ID 570568, 12 pages, http://dx.doi.org/10.1155/2015/570568

Relevant Projects

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Projectnumber	Titel	From – To
838815 (FFG)	Necrupro – Next Crude Oil Production	Mar 2013 - Mar 2014
ECO/09/256149 (FP7)	Phobior – Photobioreactor Demonstration Plant	Oct 2010 - Sep 2013
GA 2013-607-607 (Marie Curie Action)	ITN Calipso - Calcium- and light signals in photosynthetic organisms	Oct 2013 – Sep 2017

Infrastructure/Equipment

At eparella we run photobioreactors (PBR) at different scales from 40l to 5000l with a total volume of approx. 20.000l. In these setups we can cultivate various microalgae/cyanobacterial species at various environmental conditions (15-45°C, pH 2-10). Biomass harvesting and drying facilities are available using different technologies like centrifuges (GEA) or cross flow filtration setups as well as a lab scale spray dryer (Büchi) and a solar belt dryer for larger volumes. Lab facilities for QC analyses using PCR machines for DNA analysis as well as photometers (Hitachi), a PAM measurement device (LI-COR) and a microscope (Zeiss) are available. A GC analyser (Perkin Elmer) is used for fatty acid profiling. A proprietary strain collection with 100 local algae strains

is available as well. In 2018 a 1ha industrial production facility with the capacity of generating up to 100tons of algal biomass p.a. will start its production.

Profile of Main Persons

Mag. Dr. Silvia Fluch, MSc (Female)

Chief Operating Officer (COO)

Master in biology from the University of Vienna and Graz. Master of Science in organisational development at the University of Klagenfurt. Doctorate in biology at the University of Vienna, with main focus on molecular biology, marker development. Over 20 years of experience in population genetic analyses, algae research and bioinformatics. Engagement at AIT as researcher and project manager followed by team leader and senior scientist at AIT- Austrian Institute of Technology GmbH with management and budget responsibility for her unit of up to 18 people. Developer of several patents on behalf of AIT. Since 03/2015 Managing Director and chief Scientist at the ecoduna produktions-GmbH, since 09/2017 COO at eparella GmbH.

Lukas Neuwirth, MSc (Male)

Studies of Biotechnology at Universität für Bodenkultur with focus on Bio-process engineering. Master thesis on filtration technologies for micro algae. Since 2013 employed at ecoduna as process engineer dealing mostly with improving harvesting technologies. Since 2014 junior project manager. In 2017 project manager for the constructin of the 1ha industrial production site.

Andre Orfao, MSc (Male)

Studies of technological process engineering at University of Lisbon, Master thesis in WUR Wageningen on microalgae cultivation and its dependency on light regime (Cycadian clock). Since 11/2013 employed at ecoduna as process engineer with focus on biomass production and optimization of cultivation methods for maximizing output under natural light conditions. Since 04/2017 responsible process manager at ecoduna/eparella.

Daniel Kiesenhofer PhD (Male)

Education in information technology, followed by studies of biology/genetics at the University of Vienna and TU Vienna. Working at ecoduna/eparella sice 05/2017 as biology project manager.

Florian Scheweder (Male)

Studies in technical mathematics. Working at ecoduna/eparella since 2013 in the area of technical developments. Since 2017 project manager of technical R&D Projects.

4.2. Third parties involved in the project (including use of third party resources)

No third parties involved.

Does the participant plan to subcontract certain tasks (please note that core tasks of the action should not be sub-contracted)	Y
PNO will subcontract the production of the VALUEMAG website for an amount of 10.000 to the company Innovation Engineering srl (http://www.innovationengineering.eu/) who has extensive experience in developing user friendly portals, with a special focus on EU funded project ones.	

Does the participant envisage that part of its work is performed by linked third parties[1]	Y
PNO INNOVATION NBV (Belgium) to PNO Consultants S.A.S (France) and PNO NL (Netherlands): In performing the activities related to the project, PNO Innovation will make use of the expertise and infrastructures available within PNO Group, in particular will integrate in the project team the expertise available from our Innovation team in Netherlands and France. An already signed undertaking between the PNO entities specify the possibility for PNO Innovation to make use, during the project, of the high level NLbased and FR-based specialists with extensive experience in the tasks specified in the project.	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	N
If yes, please describe the third party and their contributions	

5. Ethics and security

5.1. Ethics

There are no ethical issues related to the present proposal.

5.2. Security

No activities or results raising security issues

No 'EU-classified information' as background or results are involved in the project.

[1] A third party that is an affiliated entity or has a legal link to a participant implying a collaboration not limited to the action (Article 14 of the Model Grant Agreement).

European Commission

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